

claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\agent-accde0ef1f06ab6a4.jsonl`
- SHA-256: `2f3635df84be984ee52f0276fe16d8e862b5435f135652a4a22358cc094fe78a`
- Source modified: `2026-06-14T10:22:49+00:00`
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- project: `subagents`
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`

Transcript

1. user (2026-06-14T10:09:48.239Z)

Implement AND merge the Gate-A default-flip PR (owner-approved). Repo cwd
C:/Users/avidu/Projects/badminton-highlight-indexer. ISOLATED worktree: `git -C ... fetch origin` then `git -C ... worktree add C:/Users/avidu/Projects/bhi-gateA -b feat/gate-a-default origin/master`.

GOAL: ship the validated Gate-A held-shuttle guard (`net_crossings >= 2`) as the DEFAULT detection behavior, **config-REVERSIBLE** (rollback = flip a flag). Measured win: ?F1 +0.074, CI [+0.042,+0.105], 6/6 golden, p=0.031, over-seg 1.68->1.01; corroborated on the real GX010125 reel (45->27 rallies). Gate-B stays OFF.

STEP 1 ? determine the MINIMAL, SAFE, reversible change. Read backend/config/models.py (FusionConfig.min_crossings, IndexingConfig.default_segmenter), backend/pipeline/segmenters/fusion_hybrid.py (how Gate-A applies in `\_select\_for\_handoff`), backend/main.py + config.json (served default-segmenter selection), docs/QUALITY_ITERATIONS.md (the Gate-A entry). Figure out the exact minimal change to make Gate-A apply by default while reversible. If shipping it as default requires making an UNVERIFIED served path (FusionSegmenter) the default, FLAG it and pick the safest reversible path; note if a served golden regression is warranted.

STEP 2 ? make the change + COMPREHENSIVE tests (owner wants healthy coverage + MORE CI tests). Assert: (a) the default now applies Gate-A; (b) a held-shuttle window (`net_crossings < 2`) is DROPPED by default; (c) a real rally (`>= 2`) is KEPT; (d) reversibility (`min_crossings = 0` restores prior behavior); (e) the persisted candidate SUPERSET is unchanged (Gate-A affects only the handoff). Extend test_fusion_hybrid.py / test_config.py + add new tests. ruff + mypy + full pytest pass.

STEP 3 ? RECORD the promotion: append a Gate-A promotion record to eval_baselines/experiment_leaderboard.json AND extend the docs/QUALITY_ITERATIONS.md Gate-A entry to PROMOTED-as-default with the served-flip detail. (Only YOU touch these two files; the ablation agent will not.)

STEP 4 ? GPU + Gemini are AUTHORIZED: if a golden served regression adds real confidence and is feasible, run it to confirm no regression; else flag as optional owner-gated follow-up. Do NOT block the merge on a heavy multi-video Gemini sweep ? the offline ablation (6/6, p=0.031) + unit tests are the gate.

MERGE on green (owner approved): `gh pr create --base master`; verify ALL 4 CI lanes pass; `gh pr merge <n> --squash --delete-branch`; remove the worktree + delete the branch + `git worktree prune`. If CI fails or you found a real served-verification BLOCKER, do NOT merge ? leave the PR open + report. Branch-safe (explicit paths; verify `git log origin/master..HEAD` is only your commit). End commit msg with `Co-Authored-By: Claude Opus 4.8 (1M context)`
<noreply@anthropic.com>`.

Return: PR number/URL, the exact change, the test list + results, whether you ran a regression, merge status, and any flagged risk.

2. assistant (2026-06-14T10:09:51.738Z)

I'll start by ramping up on the current state and reading the relevant files before making any changes.

3. user (2026-06-14T10:09:54.260Z)

(Bash completed with no output)

4. user (2026-06-14T10:09:54.667Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-451 of 1851 total (87024 tokens, cap 25000). Call Read with offset=452 limit=451 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
4      only on request). What just landed, what's next, open decisions."
5      metadata:
6          node_type: memory
7          type: project
8          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
9      ---
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-14 (PERSONALIZATION Phase-0 LANDED + quality checkpoint).
13     **Owner adopted the HYBRID** ? personalization as a
14     flag-gated FEATURE on a generalization-first baseline + a **contribution flywheel** ("a
15     tool that gets better the more you help it";
16     free-tier videos pool into the owner-trained baseline). Pivot analysis
17     (`scratch/PERSONALIZATION_PIVOT_ANALYSIS.md`) recommended
18     hybrid-not-personalization-first; the owner's contribution-flywheel closes the flywheel-
19     loss objection. **MERGED this session:**
20     telemetry wire-in **#140** ; per-user promotion gate **#141** (min-N floor + structural-
21     guard exemption); held-out-USER learning
22     curve **#142** ("<K videos to superior quality" = falsifiable); per-user no-regression
23     guard **#143** ; **`docs/PERSONALIZATION_PLAN.md`
24     #144** (the checked-in project doc + locked decisions). Gemini-training-source concern =
25     **CLEAN** (eval-only, gitignored, M0.3 owner-gated).
26     **LOCKED owner decisions:** identity = north-star-design/current-impl (nullable user_id,
27     device UUID); inline classifier_json behind a
28     cloud-load interface; **`min_n` is a PROMOTION-confidence floor, NOT a service gate** (a
29     1-video user is always fully segmented by the
30     baseline); free-tier video pooling (consent + minors-exclusion); **Gate-A-only
31     structural + `min_n=5`**. ?? **IN FLIGHT (3 bg agents):**
32     quality-iterations doc PR; **Gate-A/B golden-set numbers + flip-implications**
33     (`scratch/GATE_AB_NUMBERS_AND_IMPLICATIONS.md` ? leave-one-
34     signal-out ablation; owner: NUMBERS INFORM, FOUNDATIONS DECIDE); **ablation-layer
35     framework plan** (`scratch/ABLATION_FRAMEWORK_PLAN.md` ?
36     leave-one-signal-out + signal-relevance ledger over time, for the paper). **GPU + Gemini
37     API now AUTHORIZED for evals** (cooler boost on).
38     **Gate-A default flip = NOT done** ? gather numbers + implications first, then owner
39     decides (data-influenced, foundations-led). NEXT
40     personalization = Phase 1 (serving bridge: wire `LogisticRegression.load` into
41     `_select_for_handoff`) ? owner-gated, see [[weak-signals-retained]] +
42     `docs/PERSONALIZATION_PLAN.md`.
43
44     **Checkpoint:** 2026-06-14 (FUSION P3/P4 MEASURED + extraction COMPLETE + telemetry #134
45     merged + Cooler Boost). **THE NUMBERS
46     (honest LOVO n=6, IoU?0.5, C1 bootstrap-CI + sign-test):** shuttle-only F1=**0.307** ;
47     **+person (P3) ?F1=?0.021** (?6.7%, CI
48     [?0.165,+0.162], sign 2W/4L p=0.688, **CI-clears-0: NO**) ? NO lift, stays default-OFF
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(no court polygons ? `players_in_court`
31 counts spectators; per-video wash: testlarge +0.40 / gbaaddy ?0.22). **person+audio (P4
incremental) ?F1=+0.079** (+27.7%, F1
32 0.286? **0.366**, CI [?0.054,+0.203], sign 4W/2L p=0.688, **CI-clears-0: NO**) ?
PROMISING (wins 4/6, largetest/mahadevpura
33 +0.27; endpoint 0.366 > shuttle-only 0.307) but n=6 can't confirm. **VERDICT: neither
promotable (both CIs span 0); audio is
34 the lead ? need more matches + measure audio on shuttle-only directly (drop the person
handicap).** Recorded in
35 `eval_baselines/experiment_leaderboard.json`; honest negative/uncertain results KEPT
([[paper-coauthor-aim]]). **LANDED THIS
36 SESSION:** result logged to `docs/QUALITY_ITERATIONS.md` (#137);
`experiment_leaderboard.json` tracked (#139); **weak-signals
37 audit ?** (person/audio/court-bounds all present, default-OFF, no-special-casing,
retained ? refuted=false ? [[weak-signals-retained]]);
38 7-page paper-style **quality report PDF** at
`OneDrive\Documents\Rally_Detection_Quality_Report.pdf` (gen `scratch/quality_report.py`;
39 conservative academic restyle, strict-reviewer-approved); `FusionGoldenExtract`
scheduled task **unregistered** (cleanup); tree
40 clean on `master`. ?? **NEXT (fusion track):** (1) **wire `run_telemetry` (#134) into
fusion_golden + the psi wrapper** (deferred,
41 ready, mergeable); (2) **measure audio on shuttle-only directly** (quick eval, reuses
the 6 `output/*_golden_features.json`, NO new
42 data ? audio's standalone lift without the person handicap); (3) **grow the golden
corpus via Rooraa** (owner's job ? gives power
43 to confirm the +0.079 audio lead) + **activate court polygons** (re-measure person
properly ? the ?0.021 is partly a
44 `court_polygon=None` artifact); (4) future **shuttle-dropped** cue ([[weak-signals-
retained]]). **OTHER track (per
45 [[session-handoff]] ? the stated IMMEDIATE PICKUP): EXECUTION-POLICY P3 then P4**
(owner-gated; #132/#133/#135 done). **Extraction infra** (all working): detached Scheduled Task
`FusionGoldenExtract`
46 runs `scratch/extract_until_done.ps1` (self-restart loop + in-process keep-awake + GPU-
busy gate: waits if free VRAM<3500MB) ?
47 deaths were `0xC000013A` console-control kills (NOT crash/sleep/TDR/OOM ? verified via
event log), auto-recover via 3-min
48 repetition trigger + restart-on-failure. **Cooler Boost** (MSI Creator Center) dropped
85?67°C, SM ~1100?~1300MHz, velocity
49 ~26?~50 fps (the rest is Max-Q power cap, not thermal).
50
51 **Checkpoint:** 2026-06-14 (EXECUTION-POLICY framework: design + P1 + P2 SHIPPED ? **PR
#132 (design) + #133 (P1) + #135 (P2)
52 MERGED**, self-merged on green CI). Owner asked to harden the GX010125 Gemini hang as a
**policy-driven, per-job,
53 self-scheduling** system (default WAIT_AND_RESUME, ABORT, POLL_EVERY_N_SEC?, **no master
node**) with **very high coverage**
54 (failures cause delays) and **clean poll-logging** (visible but not polluting). **#132**
= design doc `docs/EXECUTION_POLICY.md`
55 (validated the owner's architecture: per-job policy = data on the job; self-schedule via
re-enqueue; no master ? rides the
56 existing single-worker executor + `jobs` table + crash sweep; DB is the coordinator).
**#133 (P1, the actual fix)** =
57 `backend/infrastructure/execution_policy.py`: `ExecutionPolicy` + `run_bounded` (HARD
`op_timeout_sec` per-attempt ? the
58 hang-killer; retries failures AND hangs; hung attempt's daemon thread abandoned) +
`poll_until` (bounded poll loop). Clocks
59 injected ? **12 tests, 100% module coverage, instant**. Refactored `gemini.py`:
`generate_content`?`run_bounded` (a
60 never-returning generate previously blocked FOREVER = the GX010125 hang; now bounded),
processing-wait?`poll_until` (**per-poll
61 logs at DEBUG, not an INFO line every 10s** ? the no-pollution ask). 23 gemini tests
stay green. **#135 (P2, substrate)** =
62 DB migration **v3** (`jobs.policy` JSON + `next_poll_at`; existing rows survive via
ALTER) + `set_job_waiting(delay_sec)`
63 (self-schedule: park 'waiting' until due) + **`claim_due_job()`** (atomic compare-and-

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set claim of queued/due-waiting ?
64     running; concurrent-safe, no master) + `create_job(policy=)`/`get_job` policy.
**Additive, no executor change**; 10 tests;
65     version asserts bumped 2?3. ? **THIS IS THE OWNER'S REAL GPU MACHINE**
(torch+CUDA/cv2/WSL/Gemini live). ?? **REMAINING
66     (owner-gated, NOT done ? deliberately not rushed at session tail given the high-coverage
bar): P3** = wire the executor
67     (main.py) to a `claim_due_job` self-scheduling loop honoring WAIT_AND_RESUME
(release+reschedule) + **resumable AI handoff**
68     (persist per-window confirmation; resume only unconfirmed ? the GX010125 reel becomes
exhaustive across resumes, no wasted
69     Gemini spend); **P4** = apply the policy to the WSL/WASB GPU pass + ffmpeg. Each = its
own distinct PR. Ties [[paper-coauthor-aim]] indirectly (infra).
70
71     **Checkpoint:** 2026-06-14 (ACTED ON FIRST-A/B FINDINGS: Gate-A PROMOTED + learned FIXED
+ GX010125 run ? **PR #130 + #131
72     MERGED** to master `2e5892e?`; self-merged on green CI). ? **THIS IS THE OWNER'S REAL
GPU MACHINE** (torch+CUDA, cv2, WSL
73     Ubuntu, Gemini key in .env all present) ? not the dev container the older CLAUDE.md note
assumes. **#130 = learned-variant
74     FIX**: additive/default-OFF `Variant.tune_threshold` (operating point on the TRAIN fold)
+ `calibrate` (Platt/isotonic, C2);
75     validated on n=6 golden ? **mean held-out F1 0.307?0.423 (+0.116), testlarge 0.000?0.300
(un-zeroed), now BEATS CONTROL
76     0.388**; threshold-in-LOVO is the big lever. **#131 = Gate-A PROMOTED** (first cue
promoted via the harness; doc trail in
77     QUALITY_ITERATIONS ? measured win dF1 +0.074, 6/6, p=0.031; recommended
`indexing.fusion.min_crossings=2`; **live served-default
78     flip stays OWNER-GATED** until a real fusion-path run confirms ? fusion P2 is unverified
on real video) **+ new
79     `docs/PER_VIDEO_OUTPUT_LAYOUT.md`** (one dir per video; fixes the hardcoded `"output"`
joins in `trajectory_hybrid.py:~176`
80     so `--output-dir` truly isolates). **GX010125_1080p RUN (offline, no GPU):** ? key
finding ? `wasb_runner.run_predict` has
81     **NO cache-hit early-return**, so the full pipeline would RE-RUN WASB (~tens of min on
1080p) + overwrite the cached CSV; the
82     cached trajectory is reusable ONLY via offline re-score. So ran
`scratch/run_gx010125_rescore.py` = parse cached
83     `output/wasb/GX010125_1080p__89605e80ed01_wasb.csv` ? 45 candidates ? CONTROL 45 rallies
(11.0min) vs **Gate-A 27** (9.3min),
84     Gate-A **dropped 18 low-crossing windows (~1.7min)** of likely held-shuttle/feed FPs ?
isolated `output/GX010125_run/` (no
85     pollution; no golden labels for this footage ? divergence not F1). A FULL Gemini-
confirmed highlight reel = a separate
86     ~tens-of-min GPU+API run (owner kicks off). Repro:
`scratch/{run_gx010125_rescore,ab_n6_lovo,ab_eval_demo}.py`. All in
87     isolated worktrees off master (siblings concurrent). ?? NEXT (owner-gated): real fusion-
path run to confirm Gate-A served-side
88     + flip the live default; impl `PER_VIDEO_OUTPUT_LAYOUT` L1; Tier-2/3 §13 corrections;
analyzers/reconcilers A1?A5/R1?R3. Ties [[paper-coauthor-aim]].
89
90     **Checkpoint:** 2026-06-14 (FIRST GOLDEN A/B EXPERIMENT + HARNESS HONEST-REPORTING WIRED
? **PR #128 + #129 MERGED**
91     to master `f680b8d`, self-merged on green CI per [[working-agreement-merging]]; owner-
directed momentum). Ran the
92     **first real end-to-end A/B on the n=6 golden corpus, 100% OFFLINE** (re-scored
persisted WASB trajectories via
93     `distill_local.build_candidates`, no GPU ? the "persist once re-score" thesis proven).
One candidate set re-scored
94     three ways: **Gate-A (net_crossings?2) is a MEASURED WIN ? ?F1 +0.074, 6/6 matches,
sign-test p=0.031, CI
95     [+0.042,+0.104], seg-count-ratio 1.68?1.01** (over-seg nearly eliminated); the **window-
level 5-feature logistic did
96     NOT clear the bar** (?0.081, 2W/4L; testlarge zeroed = threshold/calibration artifact ?
C2 motivation; **distinct from

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97     the stronger per-frame `rally_seq_proto` 0.583** ? don't conflate). Headline: **the
framework held up ? C1 promoted one
98     variant and refused another (right call both ways); C4 surfaced over-seg tIoU hid.**
**#128** = archived posterity doc
99     `docs/archives/quality-iterations/2026-06-first-ab-experiment/` (full tables) +
archives-index + live QUALITY_ITERATIONS
100     pointer. **#129 = (b) wired C1+C4 into the CORE `experiment.compare()`**
(additive/default-ON `honest_stats=True`:
101     `honest` block = per-metric bootstrap CIs + paired ?F1 CI + exact sign test + F1@k +
segment_count_ratio; `record_experiment`
102     carries honest ?F1; existing fields byte-unchanged, `honest_stats=False` = legacy
shape). Reuses #124 helpers; complements
103     sibling **#127** which wired the same C1 into `fusion_compare` (no dup). 19 existing
experiment tests pass + 3 new;
104     ruff+mypy+CI all green. Repro `scratch/ab_n6_lovo.py` + `scratch/ab_eval_demo.py`
(machine-side; read gitignored
105     manifest/external golden). All done in **isolated worktrees off master** (siblings
merged #121-#127 concurrently). ?? NEXT
106     (owner-gated): promote Gate-A via the no-regression gate (it's a real measured win); fix
the logistic (C2 calibration +
107     threshold-inside-LOVO); Tier-2/3 §13 corrections; analyzers/reconcilers impl A1?A5/R1?R3
(none started). Ties [[paper-coauthor-aim]].
108
109     **Checkpoint:** 2026-06-14 (FUSION P3/P4 MEASUREMENT BUILT + GPU EXTRACTION RUNNING ?
master `d7b1f51`; suite 514 green).
110     The "modeling" track for the person-cue (P3) + audio (P4) fusion lift ? all default-OFF,
no-special-casing, harness-measured.
111     **Merged this run (mine): #121** fusion P3 (`backend/eval/fusion_golden.py` offline
person-feature EXTRACTOR + `fusion_compare.py`
112     LOVO driver ? reuses build_candidates/PersonDetector/fusion_features/experiment.compare,
NO new scoring); **#122** P4
113     (`backend/pipeline/detectors/audio_features.py` + `backend/eval/fusion_audio.py`
incremental audio cue); **#123** progress-logging
114     + resumability (`--fresh`); **#125** single-pass
`PersonDetector.detections_over_windows` (fixed the per-window-peek O(N)
115     pathology ? ~28s/window ? one sequential decode); **#126** `--smallest-first`; **#127**
honest ?F1 = bootstrap CI + paired
116     sign test on per-video held-out F1 (wires #124's `eval_stats.paired_delta_ci`; "never a
bare mean"). Concurrent sessions
117     merged #119 (analyzers framework), #120 (run.log), #124 (eval-honesty utils:
eval_stats/score_calibration/segmentation_metrics).
118     **? GPU EXTRACTION via DETACHED SCHEDULED TASK `FusionGoldenExtract`** (runs
`scratch/extract_until_done.ps1` ? Task Scheduler
119     owns it, OUTSIDE Claude's job object ? survives turn/context/session boundaries; in-
process keep-awake + self-restart loop until
120     6/6; clean UTF-8 logs to `output/fusion_golden.log` + `scratch/extract_wrapper.log`).
Loops `python -u -m backend.eval.fusion_golden
121     --manifest output/human_lovo_manifest.json --out-dir output --smallest-first` ? 6
`output/<name>_golden_features.json` (GITIGNORED
122     + RESUMABLE: skips any video whose JSON exists). **3/6 DONE** (testlarge, mahadevpura,
largetest); remaining
123     kushagra?gbaaddy?adarsh(last, 84k frames). **MONITOR:** JSON count + `tail
output/fusion_golden.log` + `nvidia-smi`;
124     `Get-ScheduledTask FusionGoldenExtract|Select State`. **RESTART (resumes, skips done):**
`Start-ScheduledTask -TaskName
125     FusionGoldenExtract`. **STOP:** `Stop-ScheduledTask -TaskName FusionGoldenExtract` +
`Get-Process python|Stop-Process -Force`.
126     ?? **WHY DETACHED (debugged 2026-06-14):** Claude-bg-task runs DIED TWICE silently
(largestest 61%, kushagra 6% @137s), NO Python
127     traceback AND ? verified via Windows event log ? NO sleep (Kernel-Power 42), NO TDR
(Display 4101), NO WER crash, NO RADAR-OOM
128     (no event 1003; 18GB free). Traceless vanish = external `TerminateProcess`; common
factor = both were Claude-managed bg tasks
129     (job-object kill-on-close at a context boundary fits death #1 @ the prior context
summary). Scheduled Task = immune; GPU access

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130 CONFIRMED in the interactive task session (53?64% util, 3.1GB). GPU util while running = 20?64% bursty ? batch=1 Faster-RCNN

131 latency-bound (small kernels + CPU NMS + per-frame ByteTrack), NORMAL, model IS on CUDA, NOT decode-bound; batching the forward

132 pass is the next-extraction speed lever (changes features, so NOT mid-run). ~1.5-2h total. Stale Claude bg keep-awakes stopped. **DEATH #3 + GPU-job check (2026-06-14):** died a 3RD time (gbaaddy 69%)

133 ? `LastTaskResult=0xC000013A` = STATUS_CONTROL_C_EXIT = terminated by a **console-control event / CTRL+C** (NOT crash/sleep/TDR/OOM

134 ? explains the no-trace vanish; killed the detached task too, so NOT Claude's job object). GPU-contention angle CHECKED: **NO

135 competing GPU compute job** (only desktop apps + our 1 python; 5GB free; HW is a **laptop** RTX 2070S **Max-Q**, MSI). Mitigations:

136 (1) task re-registered with **3-min repetition trigger + restart-on-failure (count 10, 1-min)** ? auto-recovers within minutes of

137 ANY death; (2) **GPU-busy gate in the ps1** (owner policy = *wait until free, auto-resume*): before each python launch, if free

138 VRAM <3500MB (another job has the GPU) it waits 30s + re-checks, then launches. ***? WHEN IT COMPLETES ? the remaining steps for

139 the NUMBERS:** (1) `python -m backend.eval.fusion\_compare --features-dir output --record` ? honest P3 LOVO ?F1 (shuttle-only

140 vs +person) w/ CI+sign-test; (2) `python -m backend.eval.fusion\_audio --augment --compare --record` ? P4 incremental audio

141 ?F1; (3) log both to `docs/QUALITY\_ITERATIONS.md` I3 (HONEST measured numbers only ? [[paper-coauthor-aim]]). Honest caveat:

142 NO court polygons ? `players\_in\_court` counts spectators (3-7 on "single-court" Mahadevpura); robust signals = motion +

143 `shuttle\_player\_colocation`. GPU = RTX 2070S, person detection on Windows torch cu124 (NO WSL). Branch protocol held through

144 heavy concurrent merges (#119-127) ? [[gotcha-shared-checkout-branch-collisions]]. On master, clean.

145 **STORAGE (this session, owner-approved "delete cache + compact WSL"):** deleted the stale **93GB GX010125 frame cache**

146 (WSL `~/clips` 95?2.3GB; WSL fs used 152?60GB, 896GB free; C: 53?72GB free) ? kept the 2.2GB GX010125 proxy. The WSL

147 `ext4.vhdx` was `--set-sparse true` + `fstrim`'d (946GB trimmed) but this WSL build (2.3.24) did NOT auto-deallocate on

148 shutdown ? the file is still **~221GB allocated on C:**. **DEFERRED to post-extraction: the physical ~150GB C: reclaim via

149 `wsl --shutdown` then elevated `diskpart` ? `select vdisk file="...\CanonicalGroupLimited.Ubuntu\_79rhkp1fndgsc\LocalState\ext4.vhdx`

150 ? `compact vdisk`** (heavy C: I/O would fight the GPU run's video reads). [[gotcha-long-runs-gpu-wsl]] · [[cache-hygiene-policy]].

151

152 **Checkpoint:** 2026-06-14 (RALLY-DETECTION FRAMEWORK doc + LITERATURE REVIEW + eval-honesty code ? **PR #119 + #124 MERGED**

153 to master via self-merge on green CI per [[working-agreement-merging]]; owner-directed). **PR #119** (`docs/ANALYZERS\_AND\_RECONCILERS.md`):

154 reframed the analyzers/reconcilers design into a **multi-scope signal-fusion framework** (frame detectors / window analyzers /

155 **session analyzers**); warm-up `SessionAnalyzer` + shuttle-state frame-detector folded in) + **\$13 = a verified multi-agent

156 literature-review** (8 patterns surveyed, ~40 citations adversarially checked, **ZERO fabrications**; author-line fixes applied).

157 **Verdict (honest):** the skeleton is STANDARD prior art ? candidate-windows+scorer = Temporal Action Localization;

158 reconciler-as-refinement = Temporal Action Segmentation post-proc; "signal not verdict" = late/score-level fusion + stacked

159 generalization (~20 yrs old, already in racket sports). **Only the report-first reconciler-as-linter bundle rated

160 `novel-combination`**; warm-up-as-soft-feature = thinnest gap. **Cite, don't claim**; terminology realigned (producer?labeling

161 function, scorer?stacked meta-learner, LOVO?leave-one-group-out, tIoU?F1@k/mAP). Ties [[paper-coauthor-aim]] (\$13 is the honest

162 related-work backbone). **PR #124** implements the **Tier-1 measurement-honesty course

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corrections C1?C4** as PURE ADDITIVE /
163     default-OFF helpers (no existing compare/LOVO/metrics behaviour changed):
`backend/eval/eval_stats.py` (C1 `mean_with_ci`/
164     `bootstrap_ci`/`paired_sign_test`/`paired_delta_ci` ? never a bare LOVO mean at n?6; C3
`out_of_fold_predictions`/`grouped_folds`
165     ? no stacking leakage), `score_calibration.py` (C2 Platt+isotonic+brier/ECE ? comparable
cue confidences), `segmentation_metrics.py`
166     (C4 F1@{10,25,50} + mAP@tIoU + segment_count_ratio ? make over-segmentation tIoU is
blind to visible). 24 tests; ruff+mypy clean;
167     all 4 CI lanes green (full-suite coverage floor held). Both PRs done in **isolated git
worktrees** (off master) to dodge the
168     recurring shared-checkout branch-flips (siblings merged #117 fusion-P2, #118, #120, #123
concurrently; **#116 closed ? its content
169     had landed via #117**). ?? NEXT (owner-gated): **wire C1?C4 into the DEFAULT harness
reporting** (`compare`/`run_regression.py`
170     emit CIs + F1@k/edit-score instead of a bare mean ? changes reported numbers, so its own
PR) + the Tier-2/3 corrections in §13
171     (global reconciler decode, cue-correlation modeling, temporal smoothing, external
ShuttleSet eval). Analyzers/reconcilers IMPL
172     (A1?A5/R1?R3) still owner-gated, none started.
173
174     **Checkpoint:** 2026-06-14 (RUN.LOG READABILITY for manual QA ? **PR #120 MERGED ?
master `6076eef`**); owner asked ? self-merged
175     on verified-green CI per [[working-agreement-merging]] (squash; remote+local branch
deleted); was branch `feat/runlog-readability`
176     off `35a50f3`; CODE, minimal/config-gated). Owner's manual-QA loop = run the tool on a
video ?
177     eyeball the rally CSV against `run.log`. Two readability changes: (1) **suppress HTTP
chatter** ? new central helper
178     `backend/utils/logging_setup.py::setup_logging(verbose_http=False)` raises
`httpx/httpcore/urllib3/google_genai/google_genai/
179     google.auth/grpc` to WARNING (NOTSET to restore), gated by NEW knob
`logging.verbose_http: false` (`LoggingConfig` on
180     `AppConfig`); our `backend.*` loggers untouched; wired into `run_process_and_stitch.py`
(at import + re-applied
181     post-config-load). (2) **end-of-run rally summary** ? new pure helper
`backend/tools/export.py::format_rally_summary` (next to
182     `segments_to_csv`/`basic_stats`): header (total rallies + total play HH:MM:SS) + table
`# start end dur(s) shot_count` from the
183     produced play_segments so log?CSV line up row-for-row; handles metadata dict-or-JSON-
string, missing shot_count?`-`; HH:MM:SS
184     via local `_fmt_hms` matching the segmenters' `_fmt_ts`; logged as ONE block at end of
`main()` (also on the all-filtered
185     early-return). Tests: `test_logging_setup.py`, `test_rally_summary.py`, + `verbose_http`
default/override in `test_config.py`.
186     Suite **477 green**, ruff (`. `) + mypy (`backend training`) clean. ? Commit-msg first
attempt mangled (PowerShell `@'?'@`
187     here-string in the *Bash* tool ? literal `@`; amended via `-F file`) ? Bash tool ?
PowerShell. **BRANCH-COLLISION CLEANUP
188     (owner asked):** after merge, LOCAL checkout left clean ? on
`master`==`origin/master`==`6076eef`, 0 ahead/behind; pruned 6 stale
189     LOCAL branches mapped to MERGED/CLOSED PRs (#92/#104/#111/#106/#102 + #116-closed-
redundant) + the stale remote-tracking refs;
190     **kept** `docs/analyzers-framework` (#119 OPEN). Working tree = only the intentional
untracked carryovers (`scratch`,
191     `artifacts/`). REMOTE still has 15 prunable branches (14 merged-PR + #116) ? **offered
to prune, owner did NOT answer the
192     question ? left untouched** (must KEEP `feat/khelacharya-shot-intelligence` #11 =
deliberately-retained sibling layer per
193     [[sibling-internship-repo]], + #119). ?? NEXT: owner does the machine-side check (run on
a real video ? eyeball the quieter log +
194     summary block; dev container can't run the GPU pipeline); remote-branch prune available
on request.
195
196     **Checkpoint:** 2026-06-14 (ANALYZERS DOC ? FRAMEWORK + paper framing ? **PR #119 OPEN,

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owner-gated, do NOT merge**; branch
197   `docs/analyzers-framework` off master `35a50f3`; docs-only). Owner fed two more ideas
this session and asked to fold them in
198   and treat the doc as a **reusable framework / research paper**, not a project patch
(ties [[paper-coauthor-aim]]). Extended
199   the on-master `docs/ANALYZERS_AND_RECONCILERS.md` (which had landed via #117) from "two
seams" into a **multi-scope
200   signal-fusion framework**: producers emit SIGNALS at **3 temporal scopes** ? frame
detectors (`RallyCue`) / window analyzers
201   / **session analyzers** ? all consumed as feature columns by the learned scorer (spine
unchanged: **no special-casing**, signal
202   ?scorer, never `if/else`). Folded in: (1) **warm-up/practice** as a new
**SessionAnalyzer** tier (whole-timeline span ? an
203   `in_warmup` membership feature the scorer DOWN-WEIGHTS, NOT a hard `if t<warmup_end:
drop`; measurable on golden TODAY since
204   warm-up is already labeled negative); (2) **shuttle-state**
(`in_air`/`racket_contact`/`inactive`) placed correctly as a
205   **frame-scope detector** (a MULTI_SIGNAL_FUSION_PLAN `RallyCue`, NOT a window analyzer)
? derive-from-existing-signals first
206   (velocity+visibility+direction-reversals+audio thwack), dedicated per-frame model ONLY
where WASB trajectory gaps bite, with
207   the per-frame-label blocker flagged. Added **$10 generality & research positioning**
(weakly-supervised temporal segmentation;
208   standard-vs-contributed) with an explicit **attribution honesty caveat**: NO fabricated
citations/results, related-work +
209   ablations + >1-sport still owed, n=6 = direction-check only ? the honest backbone the
paper aim needs. Diff = `2-seam?framework`
210   (310+/137?). ? Shared checkout had flipped to master mid-task (the sibling's collision
lesson recurred) ? re-verified branch,
211   branched fresh, normal commit (no in-flight WIP left to isolate; #117/#118 already
merged so all cross-links resolve). PR #116
212   stays CLOSED (content on master via #117). ?? NEXT (owner-gated): review #119; impl
tracks unchanged (fusion P3 LOVO, then
213   analyzers/reconcilers A1?A5 / R1?R3 per the doc's phasing ? each ONE PR, default OFF,
promote on measured LOVO lift).
214
215   **Checkpoint**: 2026-06-14 (FUSION P2 SHIPPED + collision cleanup + paper aim ? master
`35a50f3`; suite 467 green; 0 open PRs).
216   **PR #117 MERGED = `FusionSegmenter` (fusion P2, registry `fusion_hybrid`, **default-
OFF**): shuttle-primary (windows seed
217   from WASB/TrackNet) + adjudicating cues ? `net_crossings` (Gate A, always persisted,
GPU-free via rally_gate w/ a reused
218   axis estimate) + optional person cue (Gate B) persisting the ~5 `fusion_features.py`
features (pure-tested) via
219   `PersonDetector.detections_per_frame` over each window's ORIGINAL-video frame range.
Plugs in via **two IDENTITY hooks** on
220   `trajectory_hybrid` (`enrich_candidate_stats`/`select_for_handoff`) ? wasb/tracknet
byte-identical (adversarially verified).
221   Served gating optional+default-OFF (`fusion.min_crossings`/`min_players`) ? gates the
handoff while persisting the full
222   SUPERSET (offline substrate intact). Court mask = optional `fusion.court_polygon`.
**Design-first** (`fusion-p2-design`
223   workflow) ? **31-agent adversarial review** (25 findings?19 confirmed; headline = NO-
REGRESSION verified) ? fixes folded
224   (frame-dim guards, round() frame idx, rally_gate `estimate`= reuse, honest torch
docstrings, +6 tests). **PR #118 MERGED** =
225   README index fix (QUALITY_ITERATIONS + EXPERIMENT_HARNESS P1-SHIPPED). **? SHARED-
CHECKOUT COLLISION (lesson)** the
226   spawn_task DESIGN session ran in THIS checkout, committed `0f673ff` (the analyzers
design doc) + flipped the branch BETWEEN
227   my branch-point check and my `checkout -b` ? my fusion branch rode on top ? **#117's
squash absorbed the design doc**, so
228   **`docs/ANALYZERS_AND_RECONCILERS.md` is on master via #117** and **PR #116 was CLOSED
as redundant** (content intact on
229   master; #116's stale "OPEN do-not-merge" checkpoint below is superseded). Re-verify `git

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branch` AND that no sibling commit
230   landed since, before `checkout -b`. **Owner side-aim ratified: co-author a paper**
"local, cheap & efficient rally detection
231   via multiple clever ideas" ? [[paper-coauthor-aim]]; `docs/QUALITY_ITERATIONS.md` is the
ablation-ready empirical backbone
232   (honest measured results only). **Analyzers+reconcilers DESIGN on master (owner-gated,
NOT started):** injectable per-window
233   analyzers (e.g. net-hit service-fault) + a report-first reconciliation pass; spine =
signals?learned-scorer, **no
234   special-casing** (owner directive). ?? NEXT (owner-gated): **fusion P3** (person
features ? `experiment.compare` LOVO lift on
235   the 6 golden videos ? the eager-person-compute cleanup lands here) + measure the Gate-A
held-shuttle drop on GPU; then
236   analyzers/reconcilers impl; P4 audio. Earlier this session: PR #114 harness, #115
person-cue extraction (see checkpoints below).
237
238   **Checkpoint:** 2026-06-14 (DESIGN DOC: two extensibility seams ? **PR #116 OPEN, owner-
gated, do NOT merge**; docs-only,
239   branch `docs/analyzers-and-reconcilers` off master `df7d5df`). Produced
`docs/ANALYZERS_AND_RECONCILERS.md` (367 lines, a
240   peer to EXPERIMENT_HARNESS + MULTI_SIGNAL_FUSION_PLAN) + a 1-line `docs/README.md`
cross-link. **Spine (owner directive):
241   NO special-casing in the decision path** ? every analyzer/reconciler emits a SIGNAL
(value+confidence); the SCORING MODEL
242   learns to weight it (already true in miniature: `experiment.predict`/_gate_mask` treat
`net_crossings`/_players_in_court`
243   as feature columns). **Seam 1 = injectable `WindowAnalyzer`s (forward)**: a registry
(mirrors `segmenter_registry`) of pure
244   per-window cues over the already-produced per-frame-aligned signals (WASB trajectory +
`PersonDetector.detections_per_frame`
245   + `rally_gate` net); each emits an `AnalyzerResult` = confidence-scored features ?
`candidate_segments.stats`, events ?
246   `events` table (buffer-on-candidate, flush-on-confirm), optional boundary hints; worked
example = a **service-fault**
247   analyzer (shuttle halts in net region ? `service_fault` signal + `net_hit` event + END
hint); injects at the FusionSegmenter
248   `_enrich_candidate_stats` hook, flag-gated default OFF. **Seam 2 = reconciliation pass
(backward)**: a report-first "linter
249   for rally decisions" ? `ConsistencyRule`s (trailing-dead-time?truncate *(owner's misfire
fix)*, no-net-crossing?drop,
250   over-merged?split, boundary-slack?tighten) flag decision-vs-evidence conflicts;
**report-first/apply-optional, NEVER silent**;
251   expressible as a harness `Variant` transform so LOVO-measurable on the 6 golden videos
BEFORE auto-apply; idempotent, fixed
252   precedence. ? **DESIGNED AGAINST the IN-FLIGHT fusion-P2 WIP** (uncommitted in the
shared tree: `fusion_hybrid.py`,
253   `fusion_features.py`, `FusionConfig.cue_flags`, `person_detector.detections_per_frame`,
`docs/QUALITY_ITERATIONS.md`) ? that
254   WIP was kept **byte-intact** (I isolated my single README line via git plumbing; tree is
CRLF / autocrlf=true, restored
255   owner README to exact session-start bytes 3841B). **Merge-order:** the doc cross-links
QUALITY_ITERATIONS.md + the fusion
256   hook, neither on master yet ? **merge #116 after (or with) fusion-P2** (noted in the PR
body). ?? NEXT: owner reviews #116;
257   implementation is the phased A1?A3 / R1?R3 plan in the doc (each ONE PR off master,
default OFF, promoted only on measured
258   LOVO lift) ? all owner-gated, none started. Ties [[candidate-segments-finetuning]] + the
harness ([[working-agreement-merging]]).
259
260   **Checkpoint:** 2026-06-13 (RALLY-QUALITY TRACK: HARNESS + PERSON-CUE EXTRACTION SHIPPED
? **PR #114 + #115 MERGED**,
261   master `df7d5df`; suite 421?441 green; 0 open PRs). **PR #115 (fusion P1):** extracted
the torchvision person detector
262   + ByteTrack + velocity windowing out of `yolo_hybrid` into
`backend/pipeline/detectors/person_detector.py`

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263     (`PersonDetector`, a reusable cue producer); `yolo_hybrid` now ORCHESTRATES only
(chunk/prefetch/AI-handoff/DB) and
264     delegates Stage-1 to `self.person`. **No behavior change** ? proven by the unchanged
`process_video` output tests (mock
265     seam moved to `self.person`); pure detection unit tests ? new
`tests/test_person_detector.py`. person_detector is
266     mypy-CLEAN (`self.model: Any`) so it needed NO quarantine; the torchvision code that put
yolo_hybrid in quarantine left
267     it, but yolo_hybrid STAYS quarantined for PRE-EXISTING orchestration debt (implicit-
Optional process_video defaults,
268     Success|Failure ABC-override mismatch, a Future[...] annotation ? flagged in mypy.ini to
ratchet separately). ruff+mypy
269     clean, CI 4/4. Docs synced (fusion-plan P1?DONE, CODE_MAP person_detector entry +
yolo_hybrid-as-orchestration, NEXT_STEPS
270     item 2 ticked). ?? **NEXT (owner-steered):** **PR 3 = fusion P2.** The cheap, highest-
confidence, PURE-LOGIC piece =
271     **wire Gate A (`min_crossings`) into the live trajectory path**
(`trajectory_hybrid`/`wasb_hybrid`) + add the knob to
272     `IndexingConfig` ? **fixes the owner's held-shuttle false positive*, default OFF,
immediately A/B-able via the harness on
273     the 6 golden trajectories. Gate B (court-mask + player-count) is **owner-gated** on the
court-mask-source decision
274     (MULTI_SIGNAL_FUSION_PLAN §9.1: manual polygon vs auto homography ? owner). **P3 learned
fusion + the SxS-on-new-videos
275     RUN need GPU/owner-side** (persist person features on real video ? train on the 6 golden
? `compare_unlabeled(CONTROL,
276     +person)` on new footage). Two PRs this session within [[working-agreement-merging]]
self-merge authority.
277
278     **Checkpoint:** 2026-06-13 (EXPERIMENT HARNESS P1 SHIPPED ? **PR #114 MERGED**, master
`ecee786`; suite 421?440 green).
279     Picked up the rally-quality track (the desktop-agent pick-up `NEXT_STEPS` flagged; both
design docs `EXPERIMENT_HARNESS.md`
280     + `MULTI_SIGNAL_FUSION_PLAN.md` arrived via the session-start `git pull`
15e4e0e?9231af2, PR #112). Built
281     `backend/eval/experiment.py` = the thin variant/experiment layer from
`EXPERIMENT_HARNESS.md` P1 (the audit's "~80%
282     already exists" HELD ? verified file:line; pure glue over
metrics/calibration/classifier/training/regression.gt_hash/
283     gemini_refine.merge_overlaps/query_candidate_segments, **NO new scoring math**):
`Variant` (JSON recipe:
284     segmenter+config_overrides+cue_flags+Gate-A `min_crossings`+Gate-B `min_players`+learned
classifier; `spec_hash`; pinned
285     `CONTROL`), `compare` (labeled A/B, **LOVO-honest** train-on-others for learned
variants, B?A deltas + `label_set_hash`),
286     **`compare_unlabeled`** (the owner's **SxS-on-NEW-videos** ask ? agreed/a_only/b_only
via match_intervals, no GT),
287     `record_experiment` (? `eval_baselines/experiment_leaderboard.json`),
`load_video_candidates` (DB bridge). 19 tests incl.
288     the load-bearing **no-regression invariant** (all cues OFF ? byte-identical to CONTROL);
ruff+mypy clean; CI 4/4. Docs
289     synced (EXPERIMENT_HARNESS P1?DONE, CODE_MAP entry + stale regression-baseline-path fix,
NEXT_STEPS item 3 ticked). Owner
290     answered ramp-up Qs: **corpus = all 6** golden videos (he said "5";
LargeTestVideo/TestLargeVideo may be cuts of one match
291     ? kept all 6 per the manifest); **merge authority = self-merge on green then continue to
PR 2** (within [[working-agreement-merging]]).
292     ?? NEXT (this session): **PR 2 = fusion P1** person/vision-cue extraction (lift
torchvision+ByteTrack+velocity out of
293     `yolo_hybrid` ? `backend/pipeline/detectors/person_detector.py` producer; yolo_hybrid
consumes it, no behavior change ?
294     regression-safe). Then P3 learned fusion trains on the 6 golden videos +
`compare_unlabeled(CONTROL,+person)` SxS on new
295     footage (GPU/owner-side). Tracks [[candidate-segments-finetuning]] + [[collector-
indexer-bridge]].

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296

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297     **Checkpoint:** 2026-06-13 (WORKING-TREE CLEANUP ? master `15e4e0e`, tree now CLEAN).
Cleaned up the leftover
298     uncommitted bits (the cache-hygiene CODE was already MERGED as #109/#110 per the
checkpoint below ? nothing to rescue
299     there): (1) `config.json` GPU-run knobs **reverted to committed defaults** (was
wasb_hybrid/7200/8 ? gemini/1800/1;
300     values stay recorded in the cache-hygiene checkpoint below ? re-apply
`default_segmenter: wasb_hybrid` in one line if
301     that's the intended new default; user didn't answer keep-vs-revert, took the safe
recoverable path). (2) stray root
302     `arial.ttf` (unused ? shipped watermark font is committed
`backend/assets/fonts/RobotoSlab-Regular.ttf` from #103) ?
303     moved to `scratch/`. (3) genuine untracked draft `docs/STORAGE_SHARING_MODEL.md`
(190-line storage-anchored
304     monetization model, owner co-design pending) ? branched + **PR #111 OPEN (owner-gated,
docs-only ? do NOT auto-merge**);
305     supersedes COMMERCIALIZATION.md framing if ratified ? ADR then). Master tree now shows
ONLY intentional carryovers
306     (untracked `scratch/` handoffs+repro scripts, `artifacts/` Gen-0 bundle) ? could be
gitignored for a pristine
307     `git status` if desired (offered, not done). Reminder: `python -m tools.cleanup_caches
--apply` reclaims the 96.5 GB
308     the SessionStart hook flagged.
309
310     **Checkpoint:** 2026-06-13 (CACHE HYGIENE shipped ? **PR #109 + docs #110 MERGED**,
master `15e4e0e`; + the ?5 reel +
311     manual cleanup). **#110** = diligence docs follow-up (README $Cache Hygiene + knobs +
tools/ line; CODE_MAP detector
312     self-clean, tools/cleanup_caches.py entry, common-task rows; min_shot_count API-parity
note) ? docs-only, CI 4/4 green.
313     Both repos now have **0 open PRs**. After the GX010125 E2E run left ~163 GB in WSL
`~/clips`, built a full cache-hygiene system (all on master):
314     (1) **self-cleaning runners** ? `indexing.{wasb,tracknet}.keep_frames` (default false)
deletes the frame cache +
315     staged video after a SUCCESSFUL run; on failure keeps them for resume; (2)
**tools/cleanup_caches.py** dry-run-first
316     ops tool (`--apply`/`--keep-video`/`--older-than`/`--include-wasbcache`/`--summary`);
(3) **disk guard** `min_free_gb`;
317     (4) **SessionStart nudge hook** in `.claude/settings.local.json` (local, gitignored).
Suite 409?421 green**, ruff+mypy
318     clean. Policy saved ? [[cache-hygiene-policy]]. ? WSL-scan footgun: the owner's login
shell mangles `for e in ` loops
319     (loop var empty) ? use du/find arg-globbing. Also this session (earlier): merged
**#107** (API min_shot_count) + **#108**
320     (watermark Windows colon fix); ran GX010125 E2E on GPU ? **GX010125_highlights.mp4** (11
rallies >7 shots) +
321     **GX010125_highlights_min5.mp4** (24 rallies ?5, watermark now works ? validates #108)
via a stitch-only reuse script
322     (no re-inference); confirmed **Gemini got 1080p** (4K only at final stitch); delivered
an execution flowchart. Manual
323     WSL cleanup freed ~68 GB (kept GX010125's 93 GB frames + proxy per owner choice). ?
LOCAL uncommitted `config.json`:
324     default_segmenter=wasb_hybrid / min_shot_count=8 / wasb.timeout_sec=7200 from the run.
See [[gotcha-long-runs-gpu-wsl]].
325
326     **Checkpoint:** 2026-06-13 (SYNC + HANDOFF REFRESH ? read-only reconciliation by a ramp-
up session; no new code from
327     me this pass). `git pull --ff-only` = already up to date at **master `6de70aa` (#108)**.
Reconciled remote vs my last
328     session (#100): seven PRs landed from OTHER slates (NOT my work ? facts from git/gh):
**#101** configurable branding
329     watermark . **#102** thread config stitching paddings into live VideoCompiler sites .
**#103** bundle Apache-2.0
330     default watermark font + fallback log . **#104** C13 field kit (flyer/playbook/answer-
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sheet/signals) . ##105 i18n
331 design plan (Kannada-first) . ##107 enforce `stitching.min_shot_count` on
/api/compile . ##108 escape Windows
332 drive-letter colon in ffmpeg filtergraph (fixes #103 watermark on Windows). OPEN:
##106 docs/video-locality-model
333 (awaiting owner). ? UNCOMMITTED WIP IN TREE (sibling/owner ? DO NOT CLOBBER): a WSL
cache-hygiene slice in
334 progress, built on my `wslCommandMixin` ? `base.py` (`_wsl_free_gb`/`_wsl_rm_rf`),
`wasb_runner.py`+`tracknet_runner.py`
335 (`keep_frames`/`min_free_gb` config + pre-run disk guard + post-success cache delete),
`models.py` (new fields),
336 `config.json` (the GPU-run knobs: default_segmenter=wasb_hybrid, min_shot_count=8,
wasb.timeout_sec=7200), untracked
337 `arial.ttf` (watermark font) + `docs/STORAGE_SHARING_MODEL.md`. Not yet PR'd by whoever
owns it. [[session-handoff]]
338 rewritten to this state.
339
340 Checkpoint: 2026-06-13 (FIRST FULL E2E GPU RUN on a real 4K GoPro match + 2 merged
fixes).
341 Owner asked to run the indexer E2E on the GPU over `F:\GoPro Hero Black
12\KhelSutra\Badminton 12
342 June 2026\GX010125.MP4` (12.8-min 4K60, 11.5GB), highlights for rallies with >7 shots,
config-driven.
343 - DELIVERED: `output\GX010125_highlights.mp4` ? 11 rallies (shot_count 8?16) of 52
detected;
344 true 4K 3840x2160 h264 59.94fps, 2:23, 1.15GB; stitched from the 4K ORIGINAL (proxy
only for indexing).
345 wasb_hybrid: 64.8% shuttle visibility, 18 candidate windows ? 52 Gemini-confirmed
rallies. min_shot_count=8
346 correctly excluded a 7-shot rally (">7" semantics). Gemini ran clean (no 503s);
~47-min wall clock.
347 - 2 PRs merged on green CI (owner's [[working-agreement-merging]] standing
authorization):
348 ##107 `api/compile` now enforces `stitching.min_shot_count` (parity w/
run_process_and_stitch);
349 ##108 sticher `_ff_quote` escapes the Windows drive-letter colon ? fixes the on-
by-default
350 bundled-font watermark (PR #103) silently failing on Windows (`fontfile='C:/...'`
broke ffmpeg
351 drawtext parsing; also fixed absolute logo `movie=` paths). Both with regression
tests; suite 409 green.
352 - The hard-won run recipe is in [[gotcha-long-runs-gpu-wsl]]: Modern Standby killed
the first 2
353 overnight attempts (keep-awake fix); cv2 PNG extraction (~38min) is the real
bottleneck not the GPU
354 (ffmpeg pre-extract, count-exact for CFR ? skips it); 1080p-proxy-index / 4K-stitch;
per-phase
355 observability (cache manifest for WASB, handoff dir for Gemini). Reusable scripts in
`scratch/`
356 (run_gx010125.py, keep_awake.ps1, probe*.py).
357 - ? LOCAL UNCOMMITTED `config.json`: default_segmenter=wasb_hybrid, min_shot_count=8,
wasb.timeout_sec=7200
358 (set for this run; revert to gemini/1/1800 if not wanted). Watermark fix NOT applied
to the delivered
359 reel (it ran before #108); a re-stitch would hit the DB segment cache + now-working
watermark (~5min).
360
361 Checkpoint: 2026-06-12 (latest ? PAID COMPONENT REMOVED + VIDEO-LOCALITY MODEL; PR
#104 updated `9c84fea` +
362 PR #106 OPEN `85abe26`, both CI green, both AWAITING OWNER). Owner spec: (1) remove
ALL associate-pay from the
363 field kit ("something else in mind") ? flyer/playbook now "Engagement terms: discussed
individually"; honesty
364 mechanics rewored ("judged on honest capture, not positive answers"); KEPT: WTP
research questions (study subject)

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365 + the ACADEMY's Phase-2 session fee (different party ? flagged, owner may strip too).
(2) Storage/upload questions
366 answered from code: uploads ARE stored in full (`security.upload\_dir`, NO retention
policy ? BACKLOG); 10GB CANNOT
367 upload (max_upload_mb=4096 rejects at init; ?90MB sequential parts; hours on Indian
uplinks). (3) NEW
368 `docs/VIDEO\_LOCALITY\_MODEL.md` (#106): central insight = timestamps are the product,
video is dead weight (only the
369 compiler touches original bytes, cuttable where the original lives); locality ladder
L0?L4 (L2 proxy-upload reuses
370 the collector ADR-002 frame-identical contract; L1-direct = client?Gemini, zero bytes
through us; L0 = owned-model
371 north star; pilot = L4 sneakernet); **frugality frame (owner): bills affordable iff GPU
doesn't burn ? L2/L1 serving
372 is GPU-less by construction, Gemini €/hr, GPU stays on the owned 2070S, per-video cost
ceiling required**;
373 **academy-visit dividend: Phase-2 consented recordings ARE the multi-court target
corpus, show-back = live L2 dress
374 rehearsal, demo reactions = build backlog**; 10 open questions Q01?Q10 (retention, MD5
identity, Gemini consent,
375 key custody, packaging, review source, reel delivery, uplink data, volume, L3 survival);
rec lean = L2 default
376 (NOT a greenlight). NB: #106's base commit was pushed by owner/sibling mid-session
(shared checkout). ?? owner:
377 review/merge #104 + #106; decide academy-fee keep/strip; the "something else" comp
structure is his to define.
378
379 **Checkpoint:** 2026-06-12 (latest ? C13 KIT REFRAMED PRODUCT-FEEDBACK-FIRST per owner
spec + **RECRUITING FLYER
380 added; PR #104 UPDATED `985a329`, CI 4/4 green, AWAITING OWNER REVIEW** ? do NOT merge
without him). Owner spec:
381 hire someone to visit academies, talk to coaches AND students (how does it help +
improvement feedback); stretch =
382 record ? pipeline ? show the academy ITS OWN sample result; UI/hosting "ready by then".
Kit now = 4 docs by audience:
383 **`PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md`** (NEW ? candidate-facing recruiting flyer, owner
fills ?/phone + recruits) .
384 **`FIELD\_VISIT\_PLAYBOOK.md`** (RENAMED from FIELD_RESEARCH_ASSOCIATE_FLYER ? day-one
playbook: demo-after-Q6 rule,
385 adult-18+-students-only w/ coach-points-out gating, Phase-2 show-back + scripted "can we
keep using it?" no-promise
386 reply) . answer sheet (+demo-reaction/student blocks) . PMF_FIELD_SIGNALS (verdict
machinery UNCHANGED; demo/student
387 data = product feedback only). 2nd adversarial critique (`w5aexakz2`, 30 agents): 12
confirmed+fixed (key honesty
388 fixes: "web app IS opening"? "we're building?will be able to try"; "match report"? "simple
summary"; "finds EVERY
389 rally" dropped; minors loophole in "senior/adult students" closed), 15 refuted. ??
owner: review #104 (thresholds
390 \$3 + flyer copy) ? merge ? fill ?/phone ? trim 60s demo reel ? recruit.
391 Owner ratified in-chat: **PMF validation = physically visiting Bangalore academies and
reading signals.** Built the
392 decision layer on top of the existing `docs/FIELD\_RESEARCH\_ASSOCIATE\_FLYER.md` (which IS
the "recruit hand-out" the
393 owner remembered; the khelacharya PDFs in OneDrive are the separate internship flyer ?
see
394 [[sibling-internship-repo]]): **`docs/PMF\_FIELD\_SIGNALS.md`** (H1-H7 ? script-question
map, G/A/R anchors, H5
395 counting rule = pre-ladder-or-budget-converted ??300 only ? anchored ?300 NEVER counts,
pinned stopping rule n=15
396 target/logistics-only/verdict-once, numeric verdicts PROCEED / PIVOT A record+index /
PIVOT B partner-API
397 (PB VisionxPodPlay footnote now in-repo) / PIVOT C parent-payer / FALSIFIED) +
`docs/FIELD\_VISIT\_ANSWER\_SHEET.md`
398 (printable per-visit form: verbatim answers, required most-negative quote, before/after-

ladder tick, OFFICE-USE

399 score grid) + flyer Q5 rewritten open-ended-first (old wording read the ?100/300/500 ladder ALOUD ? center-bias

400 manufactures the exact ?300 PROCEED threshold; caught by the 38-agent adversarial critique `ww0rvaqf3`: 1 high/2 med/

401 2 low confirmed+fixed, 9 refuted) + Stupa added to listen-for + NEXT_STEPS item 5 = kit COMPLETE. ?? owner: review

402 #104 thresholds ? merge ? fill flyer ?/phone placeholders ? recruit the associate ? first 2 visits paired. PDFs of

403 flyer+answer-sheet: generate AFTER placeholders are filled (offer stands).

404

405 ****Checkpoint:**** 2026-06-12 (latest ? ROORA VENDOR LOOP COMPLETE: ****collector #23+#24+#25 MERGED**** sequentially on

406 green CI, owner pre-authorized; master `ca61405`, 0 open PRs, only owner's `.gitignore` edit uncommitted).

407 ****#23 boundary_grid_snap BLOCKER**** in detect_issues (gcd-of-boundary-frame-diffs > ±0.3s bar ? gate BLOCKED; skips

408 CSV-path/no-fps/<3 rallies) ? ? ****replayed against the REAL 6-June PB+TT pilot zips: both auto-block with "All 12**

409 rally boundaries sit on a 30-frame grid (~0.50s @ 59.94fps)"** (the manual forensics, now automated) + ROORA_BRIEF

410 Phase-B revision (\$7 REQUIRED method: CVAT step 1, first/last box on exact contact/dead frame, sparse interior OK;

411 \$4 grid-snap auto-detected warning; \$2 skip-rallies-in-progress-at-file-edges; \$1 annotation-optimized-copy note).

412 ****#24 prep-annotation-batch****: mirrors `//` ? proxies-only tree w/ same layout

413 (thin orchestrator over the hardened single flow; per-video isolation; resumable "have" skips; live 2-court E2E

414 smoke green). ****#25 ROORA_RUNBOOK.md**** (docs/annotation/): operator E2E ? send checklist (layout?batch?manifest

415 commit?intake+email w/ the 2 non-negotiables), receive (gate decision table incl. grid-snap, review sheet,

416 import/export on ORIGINALS), troubleshooting table for every guard. Suite ****180 green****. ?? Vendor loop is now fully

417 tooled+documented. ?? AGREED NEXT (owner, session wrap 2026-06-12): (1) owner fills the canonical doc's TODO(avidullu)

418 markers + screenshots, labels it ****v8****; (2) ****joint TEST RUN** of the full vendor loop on real footage BEFORE

419 anything goes to Roora** (runbook \$1 send-side end-to-end: batch layout ? prep-annotation-batch ? manifest ?

420 mock-receive via validate-delivery on an old delivery); (3) only then send. ****v7 ANOMALY RESOLVED (2026-06-12):****

421 the canonical vendor-facing Brief+Guide is the Google Doc at docs.google.com/document/d/1NvAXE0q7t4Z0QddLkEfnta0DU982yDZaqkGNPMe8EY0

422 (provenance: generated from master `ca61405`; supersedes Drive v3/v5/v6/v7; owner confirmed superset-of-v7) ? see

423 [[roora-guidelines-drive-doc]]; before sending: pick ONE version label + fill its TODO(avidullu) markers/screenshots.

424

425 ****Checkpoint:**** 2026-06-12 (earlier ? ADR-002 IMPLEMENTED: ****collector #22 MERGED**** `a068fd1`, owner authorized

426 merge-on-green; #21 ADR + #20 + indexer #102/#103 all merged earlier today, slate was clean). Shipped: `curate

427 prep-annotation-proxy` + `src/core/proxy.py` ? ?1080p/-g 15`/CRF 19/audio-kept re-encode w/ `--fps\_mode passthrough`

428 (NEVER `--r`), post-encode parity verify (exact frame count; fps ±0.1%; duration like-with-like stream/container),

429 delete-on-ANY-failure (incl. Ctrl-C), VFR refusal w/o `--allow-vfr`, provenance ?

430 `data/annotation\_proxy\_manifest.csv` (source+proxy MD5s; video_id collision guard BEFORE encode; Excel-lock prints

431 row + keeps proxy), same-file guard normcase/realpath/samefile (NTFS case-variant `--out`+`--force` would have let

432 `ffmpeg -y` TRUNCATE the source ? review verifier reproduced it live pre-fix). Suite ****169 green**** (+32), CI green.

433 Verified live on ffmpeg 8.1.1 (smoke clip 59.94fps: parity 179 frames both files; smoke caught ``-vsync`` deprecation ?

434 ``-fps_mode``). Process: ultracode adversarial review workflow (``wyv8524zf``, 18 agents) confirmed 1 high/2 med/7 low ?

435 all fixed pre-merge. Docs: INGESTION_PIPELINE step 0 + diagram leg + onboarding FAQ + sampling-FAQ split

436 (highlights-vs-golden), ADR-002 provenance bullet as-implemented amendment, PIPELINE_ROLE ADR refs qualified.

437 ?? REMAINING ADR-002 follow-ups (separate items, NOT started): Phase-B brief revision (frame step 1 / exact boundary

438 frames / "we supply proxies") ? owner words vendor comms; grid-snap QA check in validate-delivery. First real use:

439 run prep-annotation-proxy on the next Roora batch before handoff.

440

441 ****Checkpoint:**** 2026-06-12 (earlier ? ANNOTATION-PROXY DECISION RATIFIED ? ****collector PR #21 OPEN****, docs-only

442 ****ADR-002**** in collector ``docs/DECISIONS.md``). Owner agreed in-chat w/ my rec from the assessment checkpoint below:

443 proxies generated in-repo via a future ``curate prep-annotation-proxy``. ADR records the proxy contract (SAME fps ?

444 never reduce; no trim; frame-count parity check; 1080p/CRF18-20/-g15/audio kept; intake-manifest row w/ original+proxy

445 MD5s), the why-collector rationale (owns the vendor flow = command #0 of `convert-cvat?validate-delivery?import-local;`

446 verification+provenance is the durable 80%, one-off scripts skip it), and the ****Phase-B necessity argument the owner**

447 wanted captured: ~12-15 videos/sport × ~30 min 4K/59.94 = >100k frames/file ? vendor frame-step-1 annotation

448 infeasible without proxies ? near-precondition, not optimization******. Follow-ups recorded IN the ADR: Phase-B brief

449 convention line (step=1, first/last box on exact contact/dead frame) + re-add grid-snap QA to validate-delivery.

450 Collector checkout back on master (owner's stray ``gitignore`` edit untouched). ? sibling PR #20 (below) edits

451 rally_cvat.py docstring rationale 0.5/1.0?0.5/0.5 ? no file overlap w/ #21, both docs-true. ?? owner reviews/merges

5. assistant (2026-06-14T10:09:59.686Z)

The current-state confirms the context: Gate-A was promoted in #131 but the ****live served-default flip stays owner-gated****. The owner has now approved the flip. Let me read the key files before making changes.

6. user (2026-06-14T10:10:00.641Z)

```
1 """Pydantic models for the application configuration.
2
3 This module defines the canonical, validated, typed configuration for the
4 Badminton Highlight Indexer. Built-in defaults live here (single source of truth).
5
6 The models support graceful migration from old config.json files via extra="allow".
7 """
8
9 from __future__ import annotations
10
11 from typing import Any, List, Literal, Optional
12
13 from pydantic import BaseModel, Field, field_validator, model_validator
14
15 Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18 class ValidationRelevanceConfig(BaseModel):
19     court_green_lower: list[int] = Field(default=[35, 40, 40])
20     court_green_upper: list[int] = Field(default=[85, 255, 255])
```

```

21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         relevance: ValidationRelevanceConfig =
Field(default_factory=ValidationRelevanceConfig)
31
32
33     class TrackNetConfig(BaseModel):
34         wsl_distro: str = "Ubuntu"
35         repo_dir: str = "~/models/TrackNetV4"
36         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
37         conda_env: str = "TrackNetV4"
38         weights_path: str = ""
39         queue_length: int = 5
40         stage_in_wsl: bool = True
41         wsl_stage_dir: str = "~/clips"
42         timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
43         velocity_threshold: float = 0.02
44         # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
45         # decoded frame cache) are deleted automatically ? they are pure intermediates,
46         # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
47         # true only for debugging. On FAILURE they are always kept so a re-run can resume.
48         keep_frames: bool = False
49         # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB.
50         # 0 = disabled.
51         min_free_gb: float = 0.0
52
53
54     class WasbIndexConfig(BaseModel):
55         """Typed config for the live WASB shuttle substrate (indexing.wasb).
56
57         Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
58         these knobs actually take effect ? previously this whole block was silently dropped
59         because nested config sections defaulted to extra='ignore'.
60         """
61         wsl_distro: str = "Ubuntu"
62         repo_dir: str = "~/models/WASB-SBDT"
63         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
64         conda_env: str = "wasb"
65         weights_path: str = "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
66         sport: str = "badminton"
67         wsl_stage_dir: str = "~/clips"
68         timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
69         velocity_threshold: float = 0.02
70         # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
71         # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates,
regenerable
72         # from the source. Set true only for debugging. On FAILURE they are kept so a re-run
resumes.
73         keep_frames: bool = False
74         # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB. 0 =
disabled.
75         min_free_gb: float = 0.0
76
77
78     class FusionConfig(BaseModel):
79         """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN
P2).

```



```

80
81 Every default reproduces control behaviour: the fusion segmenter persists the
82 shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']``
is
83 true ? the person-cue features, but it does NOT gate the served handoff unless a
84 threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court
mask
85 is optional: ``court_polygon=None`` ? Gate B counts every detection (player-count-
only);
86 supplied ? off-court persons (spectators / adjacent courts) are excluded.
87 """
88 # Which trajectory substrate seeds the windows (shuttle stays primary).
89 substrate: Literal["wasb", "tracknet"] = "wasb"
90 # Windowing velocity threshold for the fusion family (the engine reads
91 # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
92 # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
93 velocity_threshold: float = 0.02
94 # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
95 # enabled ? so the default path never imports torch / touches the GPU.
96 cue_flags: dict[str, bool] = Field(default_factory=dict)
97 # Served Gate A: drop windows whose shuttle net-crossings < this from the AI
handoff.
98 # 0 = OFF (no gating; persisted candidates are always the full superset for offline
re-scoring).
99 min_crossings: int = Field(default=0, ge=0)
100 # Served Gate B: when the person cue is on, drop windows with median in-court
players
101 # < this. 0 = OFF.
102 min_players: int = Field(default=0, ge=0)
103 # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
104 court_polygon: Optional[list[list[float]]] = None
105 # Net line for the person two-sided-motion split (person features only ? the shuttle
106 # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
107 net_axis: Literal["x", "y"] = "y"
108 net_position: float = Field(default=0.5, ge=0.0, le=1.0)
109
110
111 class IndexingConfig(BaseModel):
112     default_segmenter: str = "yolo_hybrid"
113     use_gpu: bool = True
114     motion_threshold: float = 0.08
115     min_segment_duration: float = 3.0
116     dead_time_merge_gap: float = 2.0
117     chunk_duration: float = 90.0
118     chunk_overlap: float = 10.0
119     detector_model: str = "fasterrcnn_resnet50_fpn"
120     detector_score_threshold: float = 0.5
121     yolo_velocity_threshold: float = 0.15
122     yolo_frame_skip: int = 3
123     yolo_tracker: str = "bytetrack.yaml"
124     yolo_prefetch_workers: int = 2
125     min_action_window_duration: float = 2.0
126     log_yolo_candidates: bool = True
127     store_yolo_candidates: bool = True
128     ai_handoff_padding: float = 2.0
129     ai_max_workers: int = 5
130     # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
131     # chunks of a high-bitrate (4K) source blow past the provider upload cap
132     # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
133     # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
134     # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
135     # Matches gemini_refine's --clip-scale-width default.
136     ai_chunk_scale_width: int = 1280
137     # Optional debug knob: trim every video to the first N seconds before processing.
138     debug_duration: Optional[float] = None

```

```

139
140     tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
141     wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
142     fusion: FusionConfig = Field(default_factory=FusionConfig)
143
144     @field_validator("default_segmenter")
145     @classmethod
146     def segmenter_must_be_registered(cls, v: str) -> str:
147         # Registry check happens at runtime in main/segmenter selection.
148         # Here we just allow any string for forward compatibility.
149         return v
150
151     @model_validator(mode="after")
152     def _chunk_step_must_be_positive(self) -> "IndexingConfig":
153         # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-
positive
154         # step makes `while t < duration: t += step` loop forever, growing memory before
any
155         # GPU work. Reject the bad config at load time instead.
156         if self.chunk_overlap >= self.chunk_duration:
157             raise ValueError(
158                 f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration
"
159                 f"({self.chunk_duration}); otherwise chunking never advances."
160             )
161         return self
162
163
164     class OutputConfig(BaseModel):
165         default_highlights_name: str = "highlights.mp4"
166         db_name: str = "sports_indexer.db"
167         # Directory where the DB, highlight reels, and processed clips are written.
168         # The CLI's --output-dir overrides this at startup (see backend/main.py:main).
169         output_dir: str = "output"
170
171
172     class WatermarkConfig(BaseModel):
173         """Branding overlay burned into each highlight clip during the per-clip re-encode
174         (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
175         (under `stitching.watermark` in config.json) to turn it off. The text is fully
176         configurable. The concat stage stays stream-copy (-c copy)."""
177         enabled: bool = True
178         text: str = "Generated by Khelsutra.guru"
179         logo_path: str = "" # optional transparent PNG overlay; "" = no logo
180         font_path: str = "" # optional .ttf; "" = use the fontconfig `font` family
below
181         font: str = "Sans" # fontconfig family used when font_path is empty
182         font_size_ratio: float = Field(default=0.035, gt=0.0, le=0.5) # text height as a
fraction of
frame height
183         opacity: float = Field(default=0.85, ge=0.0, le=1.0)
184         box: bool = True # faint backing box behind the text for legibility
185         margin: int = Field(default=24, ge=0) # px inset from the chosen corner
186         position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = "bottom-
right"
187
188
189     class StitchingConfig(BaseModel):
190         begin_padding: float = 0.5
191         end_padding: float = 0.5
192         use_cache: bool = False
193         min_shot_count: int = 1
194         watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
195
196
197     class LoggingConfig(BaseModel):

```

```

198     """Logging knobs for the run entrypoints (see backend/utils/logging_setup.py)."""
199     # Third-party HTTP/RPC client loggers (httpx, httpcore, urllib3, google-genai,
200     # google.auth, grpc) emit one INFO line per request ? dozens per Gemini run, which
201     # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
202     # them to WARNING; set true to restore full chatter for request-flow debugging.
203     verbose_http: bool = False
204
205
206     class SecurityConfig(BaseModel):
207         # When set, /api/process video_path must resolve under this directory (hosted/tenant
208         # mode). Default None preserves the single-user local 'any local file' workflow.
209         allowed_video_root: Optional[str] = None
210
211         # --- Chunked upload (B2, PR-2) ---
212         # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
213         # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
214         # contain upload_dir or /api/process will reject the uploaded paths.
215         upload_dir: str = "output/uploads"
216         # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
217         # free-tier edge proxies cap request bodies there (HOSTING_PLAN $2).
218         max_upload_mb: int = 4096
219         max_part_mb: int = 95
220         allowed_upload_extensions: List[str] = ["mp4", "mov"]
221
222
223     class AppConfig(BaseModel):
224         """Root configuration object.
225
226         All runtime configuration should be accessed via an instance of this class
227         (or the loader) rather than raw dict lookups.
228         """
229
230         sport: Sport = "badminton"
231         ai_provider: str = "gemini"
232         # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable
233         # the alias never deprecates underneath us. Eval reports record the run date
234         # alongside.
235         ai_model: str = "gemini-flash-latest"
236
237         validation: ValidationConfig = Field(default_factory=ValidationConfig)
238         indexing: IndexingConfig = Field(default_factory=IndexingConfig)
239         output: OutputConfig = Field(default_factory=OutputConfig)
240         stitching: StitchingConfig = Field(default_factory=StitchingConfig)
241         security: SecurityConfig = Field(default_factory=SecurityConfig)
242         logging: LoggingConfig = Field(default_factory=LoggingConfig)
243
244         # Allow extra keys during transition so existing config.json files do not break.
245         model_config = {
246             "extra": "allow",
247             "validate_assignment": True,
248         }
249
250         # Temporary migration shim for legacy dict-style access (config.get("section",
251         # {}).get(...)).
252         # Will be removed once all call sites (validators, segmenters, etc.) move to typed
253         # attribute access / the typed config. A nested *section* is returned as a plain
254         # dict
255         # via model_dump, so legacy chained access keeps working ? returning the Pydantic
256         # section model instead breaks it (a model has no .get), which bit every validator
257         # on
258         # real runs. Scalar leaves are returned as-is.
259         def get(self, key: str, default: Any = None) -> Any:
260             _MISSING = object()
261             val: Any = _MISSING

```

```

258         if key in type(self).model_fields or hasattr(self, key):
259             val = getattr(self, key, _MISSING)
260         if val is _MISSING:
261             # Legacy convenience: find a leaf key inside any nested section. Sections
are
262             # discovered dynamically from the model fields, so a future top-level
section
263             # (e.g. `audio`, `jobs`) is covered automatically without editing a
hardcoded list.
264             for fname in type(self).model_fields:
265                 section = getattr(self, fname, None)
266                 if isinstance(section, BaseModel) and hasattr(section, key):
267                     val = getattr(section, key)
268                     break
269         if val is _MISSING:
270             return default
271         if isinstance(val, BaseModel):
272             return val.model_dump(mode="json")
273         return val
274
275     def __getitem__(self, key: str) -> Any:
276         val = self.get(key)
277         if val is None and key not in type(self).model_fields:
278             # Re-raise like a real dict for unknown top-level
279             raise KeyError(key)
280         return val
281
282
283     # Convenience type alias for callers that want to annotate
284     Config = AppConfig
285

```

7. user (2026-06-14T10:10:00.922Z)

```

1     """Multi-signal fusion segmenter (MULTI_SIGNAL_FUSION_PLAN P2).
2
3     Shuttle stays PRIMARY: this reuses an existing trajectory substrate (WASB by default,
4     TrackNet optional) to seed candidate windows exactly as `wasb_hybrid`/`tracknet_hybrid`
5     do ? it subclasses the same `TrajectoryHybridSegmenter` engine. It then ENRICHES each
6     shuttle-seeded window via two overridable seams the engine exposes:
7
8     - ``_enrich_candidate_stats`` ? always adds the net-crossing count (Gate A signal,
9     GPU-free, computed from the trajectory we already have); and, only when the person cue
10    is explicitly enabled (``indexing.fusion.cue_flags.person``), the person-cue features
11    (``fusion_features``), persisting them into ``candidate_segments.stats`` for the learned
12    fusion (P3). The person path lazily builds ONE `PersonDetector` and only then loads
the
13    torchvision/supervision model ? so the default (person-OFF) path never loads a
detector
14    model nor touches the GPU. (torch/cv2 themselves are imported by the registry
regardless,
15    via the pre-existing yolo_hybrid ? person_detector chain; the lazy part is the model.)
16    - ``_select_for_handoff`` ? optional served Gate A/B: when
``indexing.fusion.min_crossings``
17    (and, with the person cue, ``min_players``) are raised above 0, sub-threshold windows
are
18    dropped from the AI handoff (the held-shuttle / one-sided-feed false-positive fix).
19    Persisted candidates remain the full superset regardless, so the offline
experiment
20    harness can re-score any variant.
21
22    Defaults reproduce the substrate exactly: net_crossings/person features are persisted
but
23    nothing is gated (thresholds 0, person OFF), so `FusionSegmenter` with default config
seeds

```

```

24     and hands off identically to `wasb_hybrid`. Registered as ``fusion_hybrid``, NOT the
default
25     segmenter ? it is opt-in. Promotion to default happens only on measured LOVO lift
through the
26     experiment harness (EXPERIMENT_HARNESS.md), never silently.
27     """
28     from typing import Any, Dict, List, Optional, Tuple
29
30     from backend.pipeline.segmenters.base import segmenter_registry
31     from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
32     from backend.pipeline.detectors.base import DetectorRunner
33     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
34     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
35     from backend.pipeline.detectors import rally_gate
36     from backend.pipeline.detectors import fusion_features as ff
37
38
39     class FusionSegmenter(TrajectoryHybridSegmenter):
40         """Shuttle trajectory (primary) + net-crossing Gate A + optional person cue (Gate
B)."""
41
42         family = "fusion"
43         display_label = "Fusion"
44
45         def __init__(self, db: Any, config: Any):
46             super().__init__(db, config)
47             # Built lazily only if the person cue is enabled (no detector model loaded
otherwise).
48             self._person: Optional[Any] = None
49             # Cache the play-axis/net estimate per trajectory so rally_gate doesn't re-
estimate it
50             # over the whole trajectory once per candidate window (it depends only on
`points`).
51             self._axis_cache_key: Optional[int] = None
52             self._axis_cache: Optional[tuple] = None
53
54             @property
55             def name(self) -> str:
56                 return "fusion_hybrid"
57
58             @property
59             def description(self) -> str:
60                 return ("Multi-signal fusion: shuttle trajectory (WASB/TrackNet) seeds candidate
rallies, "
61                         "net-crossing Gate A + optional person-occupancy Gate B adjudicate them,
and the "
62                         "AI provider sets semantic boundaries.")
63
64             def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
Any]]:
65                 substrate = idx_cfg.get("fusion", {}).get("substrate", "wasb")
66                 if substrate == "tracknet":
67                     cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
68                     return TrackNetRunner(cfg), {
69                         "weights_path": cfg.weights_path,
70                         "queue_length": cfg.queue_length,
71                         "fusion_substrate": "tracknet",
72                     }
73                 wcfg = WasbConfig.from_indexing_cfg(idx_cfg)
74                 return WasbRunner(wcfg), {"weights_path": wcfg.weights_path, "fusion_substrate":
"wasb"}
75
76             def _enrich_candidate_stats(self, cw: Dict[str, Any], points: List[Any], fps: float,
77                                         frame_width: float, video_path: str,
78                                         idx_cfg: Dict[str, Any]) -> Dict[str, Any]:

```

```

79         stats = super()._enrich_candidate_stats(cw, points, fps, frame_width,
video_path, idx_cfg)
80         # Gate-A signal ? net-crossing count for this window. Pure, GPU-free, always
persisted
81         # (single service feed crosses ~once; a real rally crosses >=2x). axis/net auto-
estimated
82         # over the whole trajectory by rally_gate (camera-agnostic); reused across
windows.
83         gated = rally_gate.gate_windows(points, [(cw["start"], cw["end"])], fps,
84         min_crossings=0,
estimate=self._axis_estimate(points))
85         stats["net_crossings"] = float(gated[0].crossings) if gated else 0.0
86
87         fcfg = idx_cfg.get("fusion", {})
88         if fcfg.get("cue_flags", {}).get("person", False):
89             stats.update(self._person_features(cw, points, fps, frame_width, video_path,
fcfg))
90         return stats
91
92     def _axis_estimate(self, points: List[Any]) -> tuple:
93         """Cache rally_gate's (axis, net, span) play-axis estimate per trajectory so
it's
94         computed once per video, not once per candidate window (it depends only on
points)."""
95         key = id(points)
96         if self._axis_cache_key != key or self._axis_cache is None:
97             self._axis_cache_key = key
98             self._axis_cache = rally_gate.estimate_play_axis(points)
99         est = self._axis_cache
100         assert est is not None
101         return est
102
103     def _person_features(self, cw: Dict[str, Any], points: List[Any], fps: float,
104         frame_width: float, video_path: str,
105         fcfg: Dict[str, Any]) -> Dict[str, float]:
106         """GPU path (person cue ON): run the person detector over this window's
ORIGINAL-video
107         frame range and compute the scale/translation-invariant person features. Lazily
builds
108         ONE PersonDetector. ? real-video verification is owner-side; this wiring is
mock-tested."""
109         # Lazy import so the default (person-OFF) path never pulls torch/cv2 through
this module.
110         from backend.pipeline.detectors.person_detector import PersonDetector
111         if self._person is None:
112             self._person = PersonDetector(self.config)
113
114         frame_skip = self.config.get("indexing", {}).get("yolo_frame_skip", 3)
115         # round (not truncate): int() would map e.g. 1.999s*30 to frame 59 not 60 ? a
116         # window-edge off-by-one that drops the boundary frame from the shuttle
alignment.
117         start_f, end_f = round(cw["start"] * fps), round(cw["end"] * fps)
118         det = self._person.detections_per_frame(video_path, start_f, end_f, frame_skip)
119         fw = det["frame_width"] or frame_width
120         fh = det["frame_height"]
121         # If we can't trust the frame dims, every normalised feature would silently
collapse
122         # to ~0 ? return an explicit all-zero person-feature row instead of a half-baked
one.
123         if fw <= 0 or fh <= 0:
124             return {k: 0.0 for k in ff.PERSON_FEATURE_NAMES}
125
126         poly = fcfg.get("court_polygon")
127         polygon = [(float(p[0]), float(p[1])) for p in poly] if poly else None
128         net = (fcfg.get("net_axis", "y"), float(fcfg.get("net_position", 0.5)))

```

```

129         shuttle_in = [p for p in points if start_f <= p.frame <= end_f]
130         return ff.compute_person_features(det["frames"], shuttle_in, fw, fh,
131                                         court_polygon=polygon, net=net)
132
133     def _select_for_handoff(self, windows: List[Dict[str, Any]],
134                             window_stats: List[Dict[str, Any]],
135                             idx_cfg: Dict[str, Any]) -> List[int]:
136         """Served Gate A/B over the (already-enriched) per-window stats. Default
thresholds
137         (0) keep every window ? identical to the substrate. Persisted candidates are
138         untouched (full superset)."""
139         fcfg = idx_cfg.get("fusion", {})
140         min_crossings = int(fcfg.get("min_crossings", 0))
141         min_players = int(fcfg.get("min_players", 0))
142         person_on = bool(fcfg.get("cue_flags", {}).get("person", False))
143
144         keep: List[int] = []
145         for i, st in enumerate(window_stats):
146             if min_crossings > 0 and st.get("net_crossings", 0.0) < min_crossings:
147                 continue
148             if person_on and min_players > 0 and st.get("players_in_court", 0.0) <
min_players:
149                 continue
150             keep.append(i)
151         return keep
152
153
154     segmenter_registry.register(FusionSegmenter)
155

```

8. assistant (2026-06-14T10:10:04.814Z)

Now let me look at the trajectory_hybrid engine to understand the `\_select\_for\_handoff` and `\_enrich\_candidate\_stats` base hooks, plus how the served default segmenter is selected in main.py and config.json.

9. user (2026-06-14T10:10:05.388Z)

```

1     """Shared engine for trajectory-substrate hybrid segmenters.
2
3     tracknet_hybrid and wasb_hybrid were copy-paste twins ? CODE_MAP carried a
4     standing footgun: "any P3/P4 fix must be applied to BOTH". This base class is
5     now the single home for the whole pipeline shape:
6
7     healthcheck gate -> P2 trajectory -> candidate windows (+ persistence)
8     -> P3 AI-provider handoff over padded clips -> P4 DB population/linking
9
10    A concrete subclass supplies only its identity and runner wiring:
11    - ``family`` ? config section under ``indexing.*``, output subdir, and
12      the ``<family>-hybrid-<model>`` detector tag in metadata.
13    - ``display_label`` ? human label for log lines ("WASB", "TrackNet").
14    - ``name`` / ``description`` properties (the registry contract).
15    - ``_build_runner(idx_cfg)`` ? returns (DetectorRunner, detector-specific
16      detection_params extras such as weights_path).
17
18    This base is deliberately NOT registered; only concrete subclasses register.
19    """
20    import os
21    import time
22    import logging
23    import concurrent.futures
24    from typing import Any, Dict, List, Optional, Tuple
25
26    import cv2
27

```

```

28     from backend.pipeline.segmenters.base import BasePlaySegmenter
29     from backend.utils.video import get_video_duration, extract_video_chunk
30     from backend.sports import sport_registry
31     from backend.providers import provider_registry
32     from backend.pipeline.detectors.base import DetectorRunner
33
34     logger = logging.getLogger(__name__)
35
36
37     def _fmt_ts(seconds: float) -> str:
38         seconds = max(0, int(seconds))
39         h, rem = divmod(seconds, 3600)
40         m, s = divmod(rem, 60)
41         return f"{h:02d}:{m:02d}:{s:02d}"
42
43
44     class TrajectoryHybridSegmenter(BasePlaySegmenter):
45         """Trajectory-detector hybrid: precise candidate rallies + AI semantic
46         boundaries."""
47
48         #: config section under `indexing` (velocity_threshold lives there), the
49         #: output subdir for trajectories, and the metadata detector-tag prefix.
50         family: str = ""
51         #: human-readable label used in log lines.
52         display_label: str = ""
53
54         def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
55         Any]]:
56             """Return (runner, detector-specific detection_params extras)."""
57             raise NotImplementedError
58
59         @staticmethod
60         def _probe_fps_width(video_path: str) -> tuple:
61             """Return (fps, frame_width) for a video, with sensible fallbacks."""
62             cap = cv2.VideoCapture(video_path)
63             fps = cap.get(cv2.CAP_PROP_FPS) if cap.isOpened() else 0.0
64             width = cap.get(cv2.CAP_PROP_FRAME_WIDTH) if cap.isOpened() else 0.0
65             cap.release()
66             return (fps if fps > 0 else 30.0, width if width > 0 else 1280.0)
67
68         @staticmethod
69         def _shot_count_from(segment: Dict[str, Any], duration: float) -> int:
70             """Provider-reported exchange count, else a duration-based estimate.
71
72             One canonical implementation ? the twins had drifted (wasb relied on
73             ``int(None)`` raising into the fallback; same result, by accident).
74             """
75             shot_count = segment.get("shuttle_exchange_count")
76             if shot_count is None:
77                 return max(3, int(duration / 0.8))
78             try:
79                 return int(shot_count)
80             except (ValueError, TypeError):
81                 return max(3, int(duration / 0.8))
82
83         # --- Fusion seams (identity by default ? wasb_hybrid/tracknet_hybrid unchanged)
84         -----
85         def _enrich_candidate_stats(self, cw: Dict[str, Any], points: List[Any], fps: float,
86         frame_width: float, video_path: str,
87         idx_cfg: Dict[str, Any]) -> Dict[str, Any]:
88             """Stats dict persisted into ``candidate_segments.stats`` for one window. The
89             BASE
90             returns exactly the 4 motion keys, so the trajectory twins persist byte-
91             identical
92             rows; ``FusionSegmenter`` overrides this to add ``net_crossings`` (+ person-cue

```



```

88         features). ``points`` are the whole-video trajectory points
(TrajectoryPoint)."""
89         return {
90             "peak_velocity": cw["peak_velocity"],
91             "mean_velocity": cw["mean_velocity"],
92             "active_frame_count": cw["active_frame_count"],
93             "track_id_count": cw["track_id_count"],
94         }
95
96     def _select_for_handoff(self, windows: List[Dict[str, Any]],
97                             window_stats: List[Dict[str, Any]],
98                             idx_cfg: Dict[str, Any]) -> List[int]:
99         """Indices of the candidate windows that proceed to the AI handoff (and thus may
100         become play_segments). The BASE sends ALL windows (no gating ? twins unchanged);
101         ``FusionSegmenter`` overrides this to apply Gate A/B when thresholds are raised.
102         Persisted candidates are NOT affected ? they remain the full superset for
offline
103         re-scoring."""
104         return list(range(len(windows)))
105
106     def process_video(self, video_id: str, video_path: str, # type: ignore[override]
107                       player_pool: Optional[List[str]] = None,
108                       max_frames: Optional[int] = None) -> List[Dict[str, Any]]:
109         # override-ignore: the ABC declares the Result contract (Success|Failure)
110         # but every live segmenter still returns a bare list ? the documented
111         # deliberate-incremental gap (CODE_MAP gotchas; main.py handles both).
112         label = self.display_label
113         # --- Sport profile + AI provider wiring (identical across segmenters) ---
114         sport_name = self.config.get("sport", "badminton")
115         try:
116             sport_profile = sport_registry.get(sport_name)
117         except KeyError as e:
118             logger.error(str(e))
119             return []
120
121         idx_cfg = self.config.get("indexing", {})
122         provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider",
"gemini"))
123         model_name = self.config.get("ai_model", idx_cfg.get("ai_model",
"gemini-2.5-flash"))
124
125         api_key_env_var = f"{provider_name.upper()}_API_KEY"
126         api_key = os.environ.get(api_key_env_var)
127         if not api_key:
128             logger.error(
129                 f"{api_key_env_var} environment variable is not set! "
130                 f"To run the {provider_name} handoff, set your API key. "
131                 f"In PowerShell: $env:{api_key_env_var}=\"your_api_key_here\""
132             )
133             return []
134         try:
135             provider = provider_registry.get(provider_name, api_key=api_key,
model_name=model_name)
136         except KeyError as e:
137             logger.error(str(e))
138             return []
139
140         video_duration = get_video_duration(video_path)
141         if video_duration <= 0:
142             logger.error("Invalid video duration.")
143             return []
144         fps, frame_width = self._probe_fps_width(video_path)
145
146         # --- Runner config + windowing thresholds ---
147         runner, runner_params = self._build_runner(idx_cfg)

```

```

148
149     velocity_thresh = idx_cfg.get(self.family, {}).get("velocity_threshold", 0.02)
150     merge_gap = idx_cfg.get("dead_time_merge_gap", 2.0)
151     min_window_duration = idx_cfg.get("min_action_window_duration", 2.0)
152     handoff_padding = idx_cfg.get("ai_handoff_padding", 2.0)
153     ai_max_workers = idx_cfg.get("ai_max_workers", 5)
154     log_candidates = idx_cfg.get("log_yolo_candidates", True)
155     store_candidates = idx_cfg.get("store_yolo_candidates", True)
156
157     detection_params = {
158         "detector": self.name,
159         "velocity_thresh": velocity_thresh,
160         "merge_gap": merge_gap,
161         "min_action_window_duration": min_window_duration,
162         **runner_params,
163     }
164
165     # --- Phase 1: env healthcheck (fail fast with a clear message) ---
166     ok, msg = runner.healthcheck()
167     if not ok:
168         logger.error("%s WSL environment not ready: %s", label, msg)
169         return []
170     logger.info("%s WSL env OK: %s", label, msg)
171
172     # --- Phase 2: trajectory -> candidate rally windows ---
173     # Trajectory detectors process the whole video in one pass (they carry
174     # their own frame buffering), so unlike yolo_hybrid we don't pre-chunk.
175     t_start = time.time()
176     out_dir = os.path.join("output", self.family)
177     csv_path = runner.run_predict(video_path, out_dir, video_id=video_id)
178     if not csv_path:
179         logger.error("%s produced no trajectory; aborting.", label)
180         return []
181
182     points = runner.parse_trajectory_csv(csv_path)
183     logger.info("Parsed %d trajectory points (%d visible).",
184                 len(points), sum(1 for p in points if p.visible))
185
186     all_candidate_windows = runner.trajectory_to_action_windows(
187         points, fps=fps, frame_width=frame_width, chunk_start=0.0,
188         velocity_thresh=velocity_thresh, merge_gap=merge_gap,
189         min_window_duration=min_window_duration, chunk_index=0,
190     )
191     logger.info("Phase 2 (%s) completed in %.1fs. Candidate windows: %d",
192                 label, time.time() - t_start, len(all_candidate_windows))
193
194     if log_candidates:
195         for cw in all_candidate_windows:
196             logger.info(f"[{label} rally]
197 {_fmt_ts(cw['start'])}-{_fmt_ts(cw['end'])} "
198                        f"({cw['end'] - cw['start']:.1f}s) |
199 frames={cw['active_frame_count']} "
200                        f"peak_v={cw['peak_velocity']:.3f}
201 mean_v={cw['mean_velocity']:.3f}")
202
203     # Per-window stats via the enrichment hook ? base = the 4 motion keys; the
204     # segmenter adds net_crossings + person-cue features. Computed once, reused for
205     # persistence and the handoff gate below.
206     window_stats = [
207         self._enrich_candidate_stats(cw, points, fps, frame_width, video_path,
208                                     idx_cfg)
209         for cw in all_candidate_windows
210     ]

```

```

207
208     # Persist raw candidates for the fine-tuning dataset (same table as YOLO).
209     candidate_ids: List[Optional[int]] = [None] * len(all_candidate_windows)
210     if store_candidates:
211         for i, cw in enumerate(all_candidate_windows):
212             candidate_ids[i] = self.db.add_candidate_segment(
213                 video_id, detector=self.name,
214                 start_time=cw["start"], end_time=cw["end"],
215                 chunk_index=cw["chunk_index"], confidence=min(1.0,
216                 cw["peak_velocity"]),
217                 stats=window_stats[i], detection_params=detection_params,
218             )
219     if not all_candidate_windows:
220         return []
221
222     # --- Phase 3: AI provider handoff over padded candidate clips ---
223     # Which windows reach the handoff (base = all; fusion may apply Gate A/B).
224     # candidates above stay the full superset ? only the served play_segments are
225     # gated.
226     handoff_indices = self._select_for_handoff(all_candidate_windows, window_stats,
227     idx_cfg)
228     handoff_dir = os.path.join("output", f"chunks_{provider_name}_handoff")
229     os.makedirs(handoff_dir, exist_ok=True)
230     logger.info("Phase 3: %s validation on %d candidate windows...",
231     provider_name, len(handoff_indices))
232
233     def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
234         w_start = max(0, window["start"] - handoff_padding)
235         w_end = min(video_duration, window["end"] + handoff_padding)
236         w_dur = w_end - w_start
237         w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
238         if not extract_video_chunk(video_path, w_start, w_dur, w_path,
239         reencode=True):
240             return []
241         prompt = sport_profile.get_prompt(chunk_duration=w_dur)
242         segments, _ = provider.analyze_video(w_path, prompt, chunk_duration=w_dur,
243         label=f"Handoff {wi+1}")
244         valid = []
245         for seg in segments:
246             try:
247                 seg["start_time"] = float(seg["start_time"]) + w_start
248                 seg["end_time"] = float(seg["end_time"]) + w_start
249                 seg["candidate_id"] = candidate_ids[wi]
250                 valid.append(seg)
251             except (ValueError, TypeError, KeyError) as e:
252                 logger.warning("Dropping segment with unparseable timestamps: %s ?
253         %s", seg, e)
254         try:
255             os.remove(w_path)
256         except OSError:
257             pass
258         return valid
259
260     ai_segments: List[dict] = []
261     with concurrent.futures.ThreadPoolExecutor(max_workers=ai_max_workers) as
262     executor:
263         futures = [executor.submit(process_candidate_window, wi,
264         all_candidate_windows[wi])
265         for wi in handoff_indices]
266         for future in futures:
267             ai_segments.extend(future.result())
268     try:
269         os.rmdir(handoff_dir)

```

```

264         except OSError:
265             pass
266         logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
267 len(ai_segments))
268     # --- Phase 4: DB population + candidate linking ---
269     segments_data = []
270     for segment in ai_segments:
271         try:
272             start = float(segment["start_time"])
273             end = float(segment["end_time"])
274             dur = end - start
275             metadata = {
276                 "shot_count": self._shot_count_from(segment, dur),
277                 "shot_count_confidence": segment.get("shot_count_confidence",
278 "Unknown"),
279                 "detector": f"{self.family}-hybrid-{model_name}",
280             }
281             segment_id = self.db.add_play_segment(video_id, start, end, None, None,
282 metadata)
283             if segment_id:
284                 candidate_id = segment.get("_candidate_id")
285                 if candidate_id is not None:
286                     self.db.link_candidate_to_segment(candidate_id, segment_id)
287                 segments_data.append({
288                     "id": segment_id, "start_time": start, "end_time": end,
289                     "duration": dur, "metadata": metadata,
290                 })
291         except Exception as e:
292             logger.warning("Skipping invalid segment struct: %s", e)
293     return segments_data

```

10. assistant (2026-06-14T10:10:08.487Z)

Now let me look at how the served default segmenter is selected (main.py + config.json), and the registry to understand the wasb_hybrid relationship.

11. user (2026-06-14T10:10:09.460Z)

```

1      {
2          "sport": "badminton",
3          "ai_provider": "gemini",
4          "ai_model": "gemini-flash-latest",
5          "validation": {
6              "min_resolution_width": 1280,
7              "min_resolution_height": 720,
8              "min_fps": 29.9,
9              "min_blur_threshold": 20.0,
10             "max_scene_cuts_per_minute": 0.5,
11             "relevance": {
12                 "court_green_lower": [
13                     35,
14                     40,
15                     40
16                 ],
17                 "court_green_upper": [
18                     85,
19                     255,
20                     255
21                 ],
22                 "court_pixels_min_ratio": 0.05
23             }
24         },

```

```

25     "indexing": {
26         "default_segmenter": "gemini",
27         "use_gpu": true,
28         "motion_threshold": 0.08,
29         "min_segment_duration": 3.0,
30         "dead_time_merge_gap": 2.0,
31         "chunk_duration": 90,
32         "chunk_overlap": 10,
33         "detector_model": "fasterrcnn_resnet50_fpn",
34         "detector_score_threshold": 0.5,
35         "yolo_velocity_threshold": 0.15,
36         "yolo_frame_skip": 3,
37         "yolo_tracker": "bytetrack.yaml",
38         "yolo_prefetch_workers": 2,
39         "min_action_window_duration": 2.0,
40         "log_yolo_candidates": true,
41         "store_yolo_candidates": true,
42         "ai_handoff_padding": 2.0,
43         "ai_max_workers": 1,
44         "tracknet": {
45             "wsl_distro": "Ubuntu",
46             "repo_dir": "~/models/TrackNetV4",
47             "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
48             "conda_env": "TrackNetV4",
49             "weights_path": "",
50             "queue_length": 5,
51             "stage_in_wsl": true,
52             "wsl_stage_dir": "~/clips",
53             "timeout_sec": 1800,
54             "velocity_threshold": 0.02
55         },
56         "wasb": {
57             "wsl_distro": "Ubuntu",
58             "repo_dir": "~/models/WASB-SBDT",
59             "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
60             "conda_env": "wasb",
61             "weights_path": "~/models/WASB-
62             SBDT/pretrained_weights/wasb_badminton_best.pth.tar",
63             "sport": "badminton",
64             "wsl_stage_dir": "~/clips",
65             "timeout_sec": 1800,
66             "velocity_threshold": 0.02
67         },
68         "output": {
69             "default_highlights_name": "highlights.mp4",
70             "db_name": "sports_indexer.db"
71         },
72         "stitching": {
73             "begin_padding": 0.5,
74             "end_padding": 0.5,
75             "use_cache": false,
76             "min_shot_count": 1,
77             "watermark": {
78                 "enabled": true,
79                 "text": "Generated by Khelsutra.guru",
80                 "position": "bottom-right",
81                 "opacity": 0.85,
82                 "logo_path": "",
83                 "font_path": ""
84             }
85         }
86     }

```

```

C:/Users/avidu/Projects/badminton-highlight-indexer/backend/config/models.py:112:
default_segmenter: str = "yolo_hybrid"
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/config/models.py:144:
@field_validator("default_segmenter")
Binary file C:/Users/avidu/Projects/badminton-highlight-
indexer/backend/config/__pycache__/models.cpython-313.pyc matches
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/main.py:148:    default_seg =
getattr(getattr(global_config, "indexing", None), "default_segmenter", "motion")
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/main.py:712:    seg_name =
args.segmenter or getattr(getattr(global_config, "indexing", None), "default_segmenter",
"motion")
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/reporting/harvest.py:216:
config_default = (config.get("indexing") or {}).get("default_segmenter")
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/reporting/spec.yaml:133:    The
segmenter that ran differs from config.json's default_segmenter, and no
Binary file C:/Users/avidu/Projects/badminton-highlight-
indexer/backend/reporting/__pycache__/harvest.cpython-313.pyc matches
Binary file C:/Users/avidu/Projects/badminton-highlight-
indexer/backend/__pycache__/main.cpython-313.pyc matches

```

13. assistant (2026-06-14T10:10:13.570Z)

Now let me read main.py around the segmenter selection, and the QUALITY_ITERATIONS Gate-A entry and the experiment_leaderboard.json.

14. user (2026-06-14T10:10:14.533Z)

```

130
131     `progress` is an optional callable(stage: str) used to surface coarse job progress.
132     Raises on unexpected internal errors ? the caller records those as a job failure
133     (after this function has already restored the pre-run segments, see below).
134     """
135     notify = progress or (lambda stage: None)
136
137     notify("hashing")
138     video_id = compute_video_id(req.video_path)
139     was_reingest = db_instance.get_video(video_id) is not None
140
141     # 1. Run pluggable validators (with-context variant surfaces ffprobe_meta for the
report)
142     notify("validating")
143     logger.info(f"Running validations for {req.video_path}...")
144     val_results, val_context = registry.run_all_with_context(req.video_path,
global_config)
145     failures = [r for r in val_results if not r.passed]
146
147     # typed AppConfig: attribute access for the default segmenter (with safe fallback)
148     default_seg = getattr(getattr(global_config, "indexing", None), "default_segmenter",
"motion")
149     seg_name = req.segmenter or default_seg
150     segmenter_actual = None
151     segmentation_ran = False
152     response: Optional[Dict[str, Any]] = None
153     try:
154         if failures:
155             # add_video is an UPSERT (no cascade) ? a failed re-validation only updates
156             # status; it no longer destroys the video's prior good segments.
157             db_instance.add_video(video_id, req.video_path, "failed", [r.model_dump()
for r in val_results])
158             response = {
159                 "video_id": video_id,
160                 "status": "failed",
161                 "message": "Video failed validation checks.",
162                 "results": [r.model_dump() for r in val_results],
163             }

```

```

164         return response
165
166     # 2. Run segmenter & indexer
167     logger.info("[API] Video passed checks. Segmenting and indexing...")
168     db_instance.add_video(video_id, req.video_path, "processed", [r.model_dump() for
r in val_results])
169
170     segmenter_cls = segmenter_registry.get(seg_name)
171     if not segmenter_cls:
172         # Submit-time validation already 400s unknown names; this is the defensive
173         # in-worker path (e.g. config default pointing at an unregistered
segmenter).
174         available = list(segmenter_registry.list_available().keys())
175         response = {
176             "video_id": video_id,
177             "status": "failed",
178             "message": f"Segmenter '{seg_name}' not found. Available segmenters:
{available}",
179             "results": [r.model_dump() for r in val_results],

```

15. user (2026-06-14T10:10:15.335Z)

```

700     video_id = compute_video_id(args.video_path)
701     was_reingest = db_instance.get_video(video_id) is not None
702
703     # Validate (with-context variant surfaces ffprobe_meta for the report)
704     print("[CLI] Validating...")
705     val_results, val_context = registry.run_all_with_context(args.video_path,
global_config)
706     failures = [r for r in val_results if not r.passed]
707
708     # Save video status
709     status = "failed" if failures else "processed"
710     db_instance.add_video(video_id, args.video_path, status, [r.model_dump() for r
in val_results])
711
712     seg_name = args.segmenter or getattr(getattr(global_config, "indexing", None),
"default_segmenter", "motion")
713     segmenter_actual = None
714     segmentation_ran = False
715     try:
716         print("\n" + "="*40)
717         print("VALIDATION SUMMARY")
718         print("="*40)
719         for r in val_results:
720             status_symbol = "[PASS]" if r.passed else "[FAIL]"
721             print(f"{status_symbol} {r.validator_name}: {r.message}")
722         print("="*40 + "\n")
723
724     if failures:
725         print("[FAIL] Video failed checks. Indexing halted.")
726         return
727
728     # Index
729     segmenter_cls = segmenter_registry.get(seg_name)
730     if not segmenter_cls:
731         print(f"[FAIL] Segmenter '{seg_name}' not found.")
732         return
733
734     print(f"[CLI] Indexing play segments using segmenter: {seg_name}...")
735     segmenter_actual = seg_name
736     play_wm, cand_wm = db_instance.segment_watermarks(video_id)
737     segmenter = segmenter_cls(db_instance, global_config)
738     result = segmenter.process_video(video_id, args.video_path,
max_frames=args.max_frames)

```

16. assistant (2026-06-14T10:10:21.414Z)

Now I understand the served-default selection. The default segmenter is selected by name string (`indexing.default\_segmenter`), and Gate-A's `min\_crossings` only applies inside `FusionSegmenter.\_select\_for\_handoff`. Critically: **Gate-A only fires when the served segmenter is `fusion\_hybrid`**. The current served default is `wasb\_hybrid`/`gemini` (config.json says `gemini`), and the code default is `yolo\_hybrid`.

This is the key flag described in the prompt: "If shipping it as default requires making an UNVERIFIED served path (FusionSegmenter) the default, FLAG it." Let me read the QUALITY_ITERATIONS Gate-A entry and the leaderboard before deciding the minimal change.

17. user (2026-06-14T10:10:23.039Z)

52: promote noise and can't disable a structural guard (the proven Gate-A held-shuttle guard stays on regardless of a thin
 70: **Gate-A-only structural + `min\_n = 5`**.
 91: **Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125 corroboration** ***(2026-06-14)***
 92: **Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** **The measured win**
 95: bar. **Recommended config:** **`indexing.fusion.min\_crossings = 2`** on the fusion path (the eval-only Gate A
 100: trajectory (no GPU, isolated `output/GX010125\_run/`) ? 45 candidates ? CONTROL 45 rallies vs **Gate-A 27**;
 101: Gate-A dropped **18 low-crossing windows (~1.7 min)** of likely held-shuttle/feed FPs (no golden labels for
 111: **Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
 116: **Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test **p = 0.031**, CI
 122: Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure is the C2 (calibration)
 150: **Hypothesis:** Adjudicating shuttle-seeded windows with the net-crossing **Gate A** (**`min\_crossings?2`**)
 151: and a person-occupancy **Gate B** raises **precision** on the held-shuttle / warm-up / one-sided-feed /
 160: multi-court problem is **temporal**, and the **learned model**, not a court mask, is its answer. So Gate B
 164: **Data support for Gate A.** The same archive shows real rallies on the multi-court matches contain
 168: **Shipping shape.** A NEW `fusion` segmenter, **default OFF** (**`default\_segmenter`** unchanged); Gate A
 170: byte-identical today); the court mask is an **optional** config polygon (omit ? Gate B is
 183: **rally_gate`** estimate, honest torch-import docstrings, and 6 new tests closing the person-detector / Gate-B /
 186: Gate-A-pruned windows (a P3 redesign, would change the persisted-superset semantics); and **`fusion.velocity\_threshold`**
 234: Gate A leans on this.
 240: **Gate A already existed, just wasn't wired to production** ? the cheapest precision win is plumbing, not

18. user (2026-06-14T10:10:31.593Z)

88 **directly** (without the person handicap). Recorded in
 `eval\_baselines/experiment\_leaderboard.json`. **Negative/uncertain**
 89 results are kept on purpose ? this is the honest ablation backbone for the paper aim.
 90
 91 **Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125**
 corroboration ***(2026-06-14)***
 92 **Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** **The measured win**
 93 (**?F1 +0.074, 6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg

1.68?1.01 ? the

94 [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md))
clears the honest

95 bar. ****Recommended config:**** `indexing.fusion.min\_crossings = 2` on the fusion path (the
eval-only Gate A

96 folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the
recall-oriented ***proposal***

97 candidate set; flipping the ****live served default**** stays owner-gated until a real
fusion-path run confirms it

98 (the production fusion segmenter is P2/unverified on real video) ? so this records the
promotion + rationale,

99 not a silent default flip. ****Corroborated on real footage:**** offline re-score of
GX010125_1080p's ***cached***

100 trajectory (no GPU, isolated `output/GX010125\_run/`) ? 45 candidates ? CONTROL 45
rallies vs ****Gate-A 27****;

101 Gate-A dropped ****18 low-crossing windows (~1.7 min)**** of likely held-shuttle/feed FPS
(no golden labels for

102 this footage ? divergence, not F1).

103

104 ****Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-
indexer/pull/130)).**** The

105 first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff
artifact. Added

106 additive/default-OFF `Variant.tune\_threshold` (operating point chosen on the ****train****
fold) + `calibrate`

107 (Platt/isotonic, C2). Validated on the n=6 golden set: ****mean F1 0.307 ? 0.423**
(+0.116)**; testlarge

108 0.000 ? 0.300, and the learned variant now ****beats CONTROL (0.388)****. Threshold-in-LOVO
is the big lever;

109 calibration is available too.

110

111 ****Next:**** a real fusion-path run to confirm Gate-A served-side; per-video output layout

112 (PER_VIDEO_OUTPUT_LAYOUT.md) so each video's artifacts are
self-contained.

113

114 **### First end-to-end A/B on the golden corpus (n=6) ? ****ARCHIVED**** *(2026-06-14)***

115 First real use of the harness + C1?C4 helpers, 100% offline (re-scored persisted WASB
trajectories, no GPU).

116 ****Gate-A (net-crossings ? 2) is a measured win**** ? ?F1 +0.074, 6/6 matches, sign-test
****p = 0.031****, CI

117 [+0.042, +0.104], over-segmentation (segment-count ratio) ****1.68 ? 1.01****. The window-
level 5-feature logistic

118 ****did not clear the bar**** (?0.081, 2W/4L; a threshold/calibration artifact ? ***not*** the
stronger per-frame

119 prototype). The framework held up: ****C1 promoted one variant and refused another; C4**
surfaced over-segmentation

120 tIoU hid.**** Full record + tables ?**

121 [`archives/quality-iterations/2026-06-first-ab-experiment/`](archives/quality-
iterations/2026-06-first-ab-experiment/README.md).

122 Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure
is the C2 (calibration)

123 + threshold-in-loop task list.

124

125 **### Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF**
(2026-06-14)

126 ****Why.**** A multi-agent literature review (folded into

127 ANALYZERS_AND_RECONCILERS.md §13) confirmed the
framework's ***skeleton*** is

128 standard prior art (Temporal Action Localization/Segmentation + late/score-level fusion
? cite, don't claim),

129 and flagged that our ****measurement**** is walking into known small-N traps that would make
"promote only on

130 measured lift" promote noise or hide a real win. Four corrections, shipped as ****pure,**
additive helpers** (no

131 existing `compare`/LOVO`metrics.score` behaviour changed ? callers opt in):

```

132     - C1 ? never a single mean.** `backend/eval/eval_stats.py`:
`mean_with_ci`/`bootstrap_ci` (deterministic,
133     seeded), `paired_sign_test` (the exact distribution-free "won on enough *videos*?"
test the corpus already
134     needs), `paired_delta_ci` (B?A with a CI that must clear 0). At n?6 a lone LOVO mean
is not trustworthy.
135     - C2 ? calibrate cue confidence.** `backend/eval/score_calibration.py`: Platt +
isotonic (PAVA) calibrators
136     + `brier_score`/`expected_calibration_error`, so a CNN softmax, an audio onset, and an
integer net-crossing
137     count are comparable before the arbiter and the over-confident cue can't swamp the
rest.
138     - C3 ? honest stacking.** `eval_stats.out_of_fold_predictions`/`grouped_folds` (leave-
one-group-out by
139     match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no
leakage).
140     - C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`:
`f1_at_overlaps`
141     (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU),
`segment_count_ratio`. tIoU alone
142     is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler
exists to fix.
143
144     Status. Helpers + 24 unit tests landed (pure stdlib, run in the numpy/GPU-free
lanes); ruff + mypy clean.
145     Not yet wired into the default harness flow (wiring `compare`/`run_regression.py` to
report CIs + F1@k is
146     the follow-on, owner-gated). The discipline they encode is the methodology for every
future cue measurement ?
147     see §13/C1?C4 for the citations and the remaining (Tier-2/3) corrections.
148
149     Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*
150     Hypothesis. Adjudicating shuttle-seeded windows with the net-crossing Gate A
(`min_crossings?2`)
151     and a person-occupancy Gate B raises precision on the held-shuttle / warm-up /
one-sided-feed /
152     spectator false positives ? without hurting recall, because the shuttle stays the
primary,
153     camera-invariant seed.
154
155     Scope ? and an honesty boundary from the archive (read this).** The
156     [2026-06-multivideo-lovo](archives/quality-iterations/2026-06-multivideo-
lovo/README.md) iteration
157     already smoke-tested court-region detection as a fix for the multi-court contrast
confound** and
158     found it low-headroom: only 12?14% of dead-time shuttle detections fall outside
the foreground
159     court, so a perfect court gate lifts contrast to only ~1.6?2.2× (below the 3.6× "works"
threshold) ? the
160     multi-court problem is temporal, and the learned model, not a court mask, is its
answer. So Gate B
161     here is NOT sold as the multi-court fix. It targets a different, previously-
untested failure mode
162     (held-shuttle/warm-up/one-sided/spectator FPS). Two distinct levers; don't conflate
them.

```

19. assistant (2026-06-14T10:10:38.265Z)

Now let me read the leaderboard JSON and check the registry for `wasb\_hybrid` to understand the relationship between fusion and wasb.

20. user (2026-06-14T10:10:39.067Z)

```

1      [
2      {

```

```

3      "timestamp": "2026-06-14T06:23:51.507393+00:00",
4      "iou": 0.5,
5      "label_set_hash": "50d98ffbc370",
6      "num_videos": 6,
7      "variant_a": {
8          "name": "shuttle_only",
9          "spec_hash": "47c867c36f2f",
10         "aggregate": {
11             "precision": 0.31807603780927995,
12             "recall": 0.3101947096042718,
13             "f1": 0.3068611057861412,
14             "mean_iou": 0.5791831889148886
15         }
16     },
17     "variant_b": {
18         "name": "shuttle_person_fusion",
19         "spec_hash": "253f4709b4c4",
20         "aggregate": {
21             "precision": 0.3617228001510626,
22             "recall": 0.2708939086387243,
23             "f1": 0.2863398846047612,
24             "mean_iou": 0.5690111885714347
25         }
26     },
27     "delta": {
28         "precision": 0.04364676234178266,
29         "recall": -0.039300800965475,
30         "f1": -0.02052122118138,
31         "mean_iou": -0.01017200034345389
32     },
33     "honest_delta_f1": {
34         "mean_delta": -0.02052122118137999,
35         "ci_low": -0.16462806145542527,
36         "ci_high": 0.16224446866060574,
37         "n": 6.0,
38         "ci_excludes_zero": false,
39         "sign_test": {
40             "wins": 2.0,
41             "losses": 4.0,
42             "ties": 0.0,
43             "n_effective": 6.0,
44             "p_value": 0.6875
45         }
46     }
47 },
48 {
49     "timestamp": "2026-06-14T06:24:56.487711+00:00",
50     "iou": 0.5,
51     "label_set_hash": "50d98ffbc370",
52     "num_videos": 6,
53     "variant_a": {
54         "name": "shuttle_person",
55         "spec_hash": "ee89f1e41604",
56         "aggregate": {
57             "precision": 0.3617228001510626,
58             "recall": 0.2708939086387243,
59             "f1": 0.2863398846047612,
60             "mean_iou": 0.5690111885714347
61         }
62     },
63     "variant_b": {
64         "name": "shuttle_person_audio",
65         "spec_hash": "d68c39eba87c",
66         "aggregate": {
67             "precision": 0.4530798177857001,

```

```

68         "recall": 0.3408138120839734,
69         "f1": 0.3655826552797436,
70         "mean_iou": 0.7074802852140344
71     }
72 },
73     "delta": {
74         "precision": 0.0913570176346375,
75         "recall": 0.06991990344524907,
76         "f1": 0.07924277067498242,
77         "mean_iou": 0.13846909664259965
78     },
79     "honest_delta_f1": {
80         "mean_delta": 0.07924277067498244,
81         "ci_low": -0.053508315003560535,
82         "ci_high": 0.2029045239922802,
83         "n": 6.0,
84         "ci_excludes_zero": false,
85         "sign_test": {
86             "wins": 4.0,
87             "losses": 2.0,
88             "ties": 0.0,
89             "n_effective": 6.0,
90             "p_value": 0.6875
91         }
92     }
93 }
94 ]

```

21. assistant (2026-06-14T10:10:43.722Z)

Now let me check the registry and how `wasb\_hybrid` registers, and whether `FusionSegmenter` with default config produces byte-identical behavior to `wasb\_hybrid`. This is the crux of the safe reversible path. Let me check the registry listing and the wasb_hybrid segmenter.

22. user (2026-06-14T10:10:46.224Z)

```

23: class WasbHybridSegmenter(TrajectoryHybridSegmenter):
24:     family = "wasb"
28:     def name(self) -> str:
41: segmenter_registry.register(WasbHybridSegmenter)

```

23. user (2026-06-14T10:10:46.424Z)

```

1     """WASB-based hybrid play segmenter.
2
3     Phase 2 uses **WASB** (nttcom/WASB-SBDT, MIT) shuttle-trajectory detection via
4     WSL2 (backend/pipeline/detectors/wasb_runner.py). WASB ships pretrained
5     badminton weights and runs today, so this is the live shuttle-motion substrate.
6
7     For the per-video *calibration* goal we mainly use Phase 2 (trajectory ->
8     candidate windows); the windowing thresholds (velocity_thresh / merge_gap /
9     min_window_duration) are what get tuned against golden rally labels.
10
11     The pipeline engine (healthcheck -> trajectory -> windows -> AI handoff -> DB)
12     lives in trajectory_hybrid.TrajectoryHybridSegmenter; this class only wires the
13     WASB runner and identity.
14     """
15     from typing import Any, Dict, Tuple
16
17     from backend.pipeline.segmenters.base import segmenter_registry
18     from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
19     from backend.pipeline.detectors.base import DetectorRunner
20     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
21

```

```

22
23     class WasbHybridSegmenter(TrajectoryHybridSegmenter):
24         family = "wasb"
25         display_label = "WASB"
26
27         @property
28         def name(self) -> str:
29             return "wasb_hybrid"
30
31         @property
32         def description(self) -> str:
33             return ("Two-stage pipeline: WASB shuttle-trajectory detection (WSL2, pretrained
badminton "
34                     "weights) for precise candidate rallies + AI provider for semantic
boundaries.")
35
36         def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
Any]]:
37             cfg = WasbConfig.from_indexing_cfg(idx_cfg)
38             return WasbRunner(cfg), {"weights_path": cfg.weights_path}
39
40
41     segmenter_registry.register(WasbHybridSegmenter)
42

```

24. assistant (2026-06-14T10:10:55.884Z)

Now I have the complete picture. Let me analyze the critical design decision.

****Key insight on `family` and config sections:****

- `WasbHybridSegmenter.family = "wasb"` ? reads `idx\_cfg["wasb"]` for velocity_threshold etc. and writes output to `output/wasb`.
- `FusionSegmenter.family = "fusion"` ? reads `idx\_cfg["fusion"]` and writes output to `output/fusion`. Its `\_build\_runner` itself reads `idx\_cfg["fusion"]["substrate"]` (default "wasb") and uses `WasbConfig.from\_indexing\_cfg(idx\_cfg)` (which reads `idx\_cfg["wasb"]`). So fusion's WASB runner uses the *same* WASB config as wasb_hybrid.

So `FusionSegmenter` with default config = byte-identical seeding/handoff to `wasb\_hybrid` (the docstring states this, and #117's review verified it). Promoting Gate-A to default means ****the served default segmenter must become `fusion\_hybrid` AND `fusion.min\_crossings = 2`****.

This is exactly the flagged risk in the prompt. Let me verify the test files and the existing fusion tests before deciding the minimal change.

25. user (2026-06-14T10:10:56.603Z)

```

test_config.py
test_config_wasb.py
test_export_wasb_segments.py
test_fusion_compare.py
test_fusion_features.py
test_fusion_golden_telemetry.py
test_fusion_hybrid.py
test_segmentation_metrics.py

```

26. user (2026-06-14T10:10:56.808Z)

```

1     """FusionSegmenter tests (mocked runner/provider/db ? no GPU/WSL).
2
3     Proves: net_crossings is persisted; the person cue is OFF by default (no person-feature
4     keys, PersonDetector never built); the persisted candidate set is the full superset of
the
5     substrate windows; served Gate A drops sub-threshold windows from the AI handoff while
6     persistence keeps the superset; and the person cue, when enabled, adds the person

```

```

features.
7     """
8     from types import SimpleNamespace
9     from unittest.mock import MagicMock, patch
10
11     from backend.pipeline.detectors.fusion_features import PERSON_FEATURE_NAMES
12     from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter
13
14     # Window A [1,4]s: shuttle alternates across the net every frame -> many crossings (a
15     rally).
16     # Window B [10,12]s: shuttle sits at the net level (held) -> 0 crossings. fps=30.
17     WIN_A = {"start": 1.0, "end": 4.0, "active_frame_count": 90, "peak_velocity": 0.4,
18             "mean_velocity": 0.2, "track_id_count": 2, "chunk_index": 0}
19     WIN_B = {"start": 10.0, "end": 12.0, "active_frame_count": 60, "peak_velocity": 0.1,
20             "mean_velocity": 0.05, "track_id_count": 1, "chunk_index": 0}
21
22     def _points():
23         pts = []
24         for f in range(30, 120):          # window A frames: y flips 100<->900 around the net
25             pts.append(SimpleNamespace(frame=f, visible=True, x=500.0, y=100.0 if f % 2 == 0
26             else 900.0))
27         for f in range(300, 360):          # window B frames: y == net level -> deadband -> no
28             pts.append(SimpleNamespace(frame=f, visible=True, x=500.0, y=500.0))
29         return pts
30
31     def _runner():
32         r = MagicMock()
33         r.healthcheck.return_value = (True, "env ok")
34         r.run_predict.return_value = "out.csv"
35         r.parse_trajectory_csv.return_value = _points()
36         r.trajectory_to_action_windows.return_value = [dict(WIN_A), dict(WIN_B)]
37         return r
38
39     def _run(indexing=None, provider_segments=None):
40         provider = MagicMock()
41         provider.analyze_video.return_value = (provider_segments or [{"start_time": 0.5,
42         "end_time": 2.0}], {})
43         db = MagicMock()
44         db.add_candidate_segment.side_effect = [101, 102, 103, 104]
45         db.add_play_segment.return_value = 200
46
47         with patch.object(FusionSegmenter, "_build_runner", return_value=_runner(),
48         {"weights_path": "w"})), \
49             patch.object(FusionSegmenter, "_probe_fps_width", return_value=(30.0, 1280.0)), \
50             patch("backend.pipeline.segmenters.trajectory_hybrid.get_video_duration",
51             return_value=30.0), \
52             patch("backend.pipeline.segmenters.trajectory_hybrid.extract_video_chunk",
53             return_value=True), \
54             patch("backend.pipeline.segmenters.trajectory_hybrid.provider_registry") as
55             preg, \
56             patch("os.environ.get", return_value="dummy_key"):
57             preg.get.return_value = provider
58             seg = FusionSegmenter(db, {"indexing": indexing or {}})
59             res = seg.process_video("v1", "clip.mp4")
60             return db, provider, res
61
62     def _stats_calls(db):
63         """Persisted stats dicts, in call order."""

```

```

61         return [kw["stats"] for _a, kw in db.add_candidate_segment.call_args_list]
62
63
64     def test_net_crossings_persisted_and_person_off_by_default():
65         db, _provider, _res = _run()
66         stats = _stats_calls(db)
67         assert len(stats) == 2                                # both windows persisted
68     (superset)
69         assert all("net_crossings" in s for s in stats)
70         # Window A (rally) crosses many times; window B (held) zero.
71         assert stats[0]["net_crossings"] >= 2
72         assert stats[1]["net_crossings"] == 0.0
73         # Person cue OFF by default -> NONE of the person features present.
74         assert all(not any(k in s for k in PERSON_FEATURE_NAMES) for s in stats)
75         # The 4 base motion keys are still there.
76         assert all({"peak_velocity", "mean_velocity", "active_frame_count",
77 "track_id_count"} <= set(s)
78         for s in stats)
79
80
81     def test_persisted_candidates_are_superset_of_substrate_windows():
82         db, _provider, _res = _run()
83         spans = [(kw["start_time"], kw["end_time"]) for _a, kw in
84 db.add_candidate_segment.call_args_list]
85         assert spans == [(1.0, 4.0), (10.0, 12.0)]            # exactly the substrate windows,
86     unfiltered
87
88
89     def test_default_hands_off_all_windows():
90         _db, provider, _res = _run()                            # min_crossings default 0 -> no
91     gating
92         assert provider.analyze_video.call_count == 2
93
94
95     def test_gate_a_drops_low_crossing_window_from_handoff_only():
96         db, provider, _res = _run(indexing={"fusion": {"min_crossings": 2}})
97         # Persistence is still the full superset (offline substrate intact)...
98         assert len(_stats_calls(db)) == 2
99         # ...but only the high-crossing window A reaches the AI handoff.
100         assert provider.analyze_video.call_count == 1
101
102
103     def test_person_cue_adds_features_when_enabled():
104         # Two in-court players, each shifting between two sampled frames ? exercises the
105     REAL
106         # glue (in-court counts -> players_in_court; track displacement ->
107     mean_player_motion),
108         # not just key-presence.
109         fake_pd = MagicMock()
110         fake_pd.detections_per_frame.return_value = {
111             "frames": {
112                 30: [(1, (10, 10, 30, 40)), (2, (60, 10, 80, 40))],
113                 33: [(1, (16, 10, 36, 40)), (2, (66, 10, 86, 40))],
114             },
115             "fps": 30.0, "frame_width": 1000.0, "frame_height": 1000.0,
116         }
117         with patch("backend.pipeline.detectors.person_detector.PersonDetector",
118 return_value=fake_pd):
119             db, _provider, _res = _run(indexing={"fusion": {"cue_flags": {"person": True}}})
120             stats = _stats_calls(db)
121             assert all(all(k in s for k in PERSON_FEATURE_NAMES) for s in stats)
122             assert all("net_crossings" in s for s in stats)
123             assert stats[0]["players_in_court"] == 2.0          # 2 in-court players every
124     sampled frame
125             assert stats[0]["mean_player_motion"] > 0.0        # both tracks moved -> non-zero

```

```

motion
117         fake_pd.detections_per_frame.assert_called()           # the person detector WAS used
118
119
120     def test_gate_b_drops_low_player_window_from_handoff():
121         """Gate B: with the person cue on and min_players=2, a window whose in-court player
122         count is below 2 is dropped from the handoff ? but still persisted (superset)."""
123         def _side(video_path, start_f, end_f, frame_skip):
124             if start_f >= 300:           # window B [10,12]s -> start frame ~300: only 1 player
125                 frames = {start_f: [(1, (10, 10, 30, 40))]}
126             else:                       # window A [1,4]s -> start frame ~30: 2 players
127                 frames = {start_f: [(1, (10, 10, 30, 40)), (2, (60, 10, 80, 40))]}
128             return {"frames": frames, "fps": 30.0, "frame_width": 1000.0, "frame_height":
1000.0}
129
130         fake_pd = MagicMock()
131         fake_pd.detections_per_frame.side_effect = _side
132         with patch("backend.pipeline.detectors.person_detector.PersonDetector",
133         return_value=fake_pd):
134             db, provider, _res = _run(indexing={"fusion": {"cue_flags": {"person": True},
135         "min_players": 2}})
136             stats = _stats_calls(db)
137             assert len(stats) == 2           # superset persisted (both)
138             assert provider.analyze_video.call_count == 1       # only window A (>=2 players)
139         handed off
140             assert stats[0]["players_in_court"] >= 2 and stats[1]["players_in_court"] < 2
141
142     def test_build_runner_selects_substrate():
143         """_build_runner picks the WASB or TrackNet runner by indexing.fusion.substrate."""
144         seg = FusionSegmenter(MagicMock(), {"indexing": {}})
145         with patch("backend.pipeline.segmenters.fusion_hybrid.WasbRunner") as W, \
146             patch("backend.pipeline.segmenters.fusion_hybrid.WasbConfig") as WC, \
147             patch("backend.pipeline.segmenters.fusion_hybrid.TrackNetRunner") as T, \
148             patch("backend.pipeline.segmenters.fusion_hybrid.TrackNetConfig") as TC:
149             WC.from_indexing_cfg.return_value = MagicMock(weights_path="w")
150             TC.from_indexing_cfg.return_value = MagicMock(weights_path="t", queue_length=5)
151             _runner_w, extras_w = seg._build_runner({"fusion": {"substrate": "wasb"}})
152             _runner_t, extras_t = seg._build_runner({"fusion": {"substrate": "tracknet"}})
153             _runner_d, extras_d = seg._build_runner({})           # default -> wasb
154             assert W.called and extras_w["fusion_substrate"] == "wasb"
155             assert T.called and extras_t["fusion_substrate"] == "tracknet"
156             assert extras_t["queue_length"] == 5
157             assert extras_d["fusion_substrate"] == "wasb"

```

27. assistant (2026-06-14T10:11:00.676Z)

Excellent. The existing tests already cover Gate-A mechanics thoroughly. Now I have a complete understanding. Let me check `test\_config.py` to see the config-default test structure and confirm where to add the new default-assertion test.

28. user (2026-06-14T10:11:02.218Z)

```

15:from backend.config import load_config, save_default_config, AppConfig
18:def test_load_default_config():
21:     assert isinstance(cfg, AppConfig)
23:     assert cfg.indexing.default_segmenter == "yolo_hybrid"
28:def test_logging_config_quiets_http():
34:def test_logging_config_file_override():
43:def test_fusion_config_defaults_reproduce_control():
49:     assert f.min_crossings == 0           # Gate A OFF
55:def test_fusion_config_file_override():
60:     "substrate": "tracknet", "min_crossings": 2,
66:     assert f.min_crossings == 2

```



```

73: def test_file_override():
78:     "default_segmenter": "gemini",
89:     assert cfg.indexing.default_segmenter == "gemini"
96: def test_extra_keys_tolerated():
102:     "indexing": {"default_segmenter": "yolo_hybrid"},
110:     assert cfg.indexing.default_segmenter == "yolo_hybrid"
114: def test_unknown_keys_warn_with_dotted_paths(tmp_path, caplog):
127:     assert isinstance(cfg, AppConfig) # still non-fatal
134: def test_recognized_config_does_not_warn(tmp_path, caplog):
138:     "indexing": {"default_segmenter": "gemini", "wasb": {"velocity_threshold": 0.02}},
146: def test_save_default_config_roundtrip():
154:     assert cfg.indexing.default_segmenter == "yolo_hybrid"
158: def test_model_is_the_source_of_truth():
161:     cfg = AppConfig()
165: def test_output_dir_field_default_and_override():
169:     cfg = AppConfig()
186: def test_get_returns_dict_for_nested_sections_legacy_compat():
191:     cfg = AppConfig()
198:     assert cfg.get("indexing", {}).get("default_segmenter") == "yolo_hybrid"

```

29. user (2026-06-14T10:11:05.574Z)

```

15     from backend.config import load_config, save_default_config, AppConfig
16
17
18     def test_load_default_config():
19         # Force load without the project's config.json so we test the model built-in
defaults
20         cfg = load_config("/nonexistent/path/that/does/not/exist/12345/config.json")
21         assert isinstance(cfg, AppConfig)
22         assert cfg.sport == "badminton"
23         assert cfg.indexing.default_segmenter == "yolo_hybrid"
24         assert cfg.validation.min_blur_threshold == 20.0 # updated default from review
25         assert cfg.indexing.ai_max_workers == 5
26
27
28     def test_logging_config_default_quiets_http():
29         """verbose_http defaults to False so run.log is quiet unless explicitly turned
up."""
30         cfg = load_config("/nonexistent/path/12345/config.json")
31         assert cfg.logging.verbose_http is False
32
33
34     def test_logging_config_file_override():
35         with tempfile.TemporaryDirectory() as td:
36             p = os.path.join(td, "logging.json")
37             with open(p, "w") as fh:
38                 json.dump({"logging": {"verbose_http": True}}, fh)
39             cfg = load_config(p)
40             assert cfg.logging.verbose_http is True
41
42
43     def test_fusion_config_defaults_reproduce_control():
44         """The fusion block defaults must be no-ops: wasb substrate, no cues, no gating, no
mask."""
45         cfg = load_config("/nonexistent/path/12345/config.json")
46         f = cfg.indexing.fusion
47         assert f.substrate == "wasb"
48         assert f.cue_flags == {} # person cue OFF
49         assert f.min_crossings == 0 # Gate A OFF
50         assert f.min_players == 0 # Gate B OFF
51         assert f.court_polygon is None # no mask
52         assert f.velocity_threshold == 0.02
53
54

```

```

55     def test_fusion_config_file_override():
56         with tempfile.TemporaryDirectory() as td:
57             p = os.path.join(td, "fusion.json")
58             with open(p, "w") as fh:
59                 json.dump({"indexing": {"fusion": {
60                     "substrate": "tracknet", "min_crossings": 2,
61                     "cue_flags": {"person": True}, "court_polygon": [[0.1, 0.1], [0.9, 0.1],
[0.5, 0.9]],
62                     }}, fh)
63             cfg = load_config(p)
64             f = cfg.indexing.fusion
65             assert f.substrate == "tracknet"
66             assert f.min_crossings == 2
67             assert f.cue_flags == {"person": True}
68             assert len(f.court_polygon) == 3
69             # An old config without a fusion block still loads (defaults applied).
70             assert cfg.indexing.wasb.velocity_threshold == 0.02
71
72
73     def test_file_override():
74         with tempfile.TemporaryDirectory() as td:
75             p = os.path.join(td, "override.json")
76             data = {
77                 "indexing": {
78                     "default_segmenter": "gemini",
79                     "ai_max_workers": 3,
80                 },
81                 "validation": {
82                     "min_blur_threshold": 42.0,
83                 },
84             }
85             with open(p, "w") as f:
86                 json.dump(data, f)
87
88             cfg = load_config(p)
89             assert cfg.indexing.default_segmenter == "gemini"

```

30. assistant (2026-06-14T10:11:22.463Z)

Now I have everything I need. Let me think through the **minimal, safe, reversible change** carefully.

Design decision

What "Gate-A as default" means concretely: Gate-A (`min_crossings >= 2`) only fires inside `FusionSegmenter._select_for_handoff``. For it to apply by default, two things must be true at serving time:

1. The served segmenter must be `fusion_hybrid`` (so `_select_for_handoff`` runs the Gate-A logic).
2. `indexing.fusion.min_crossings`` must default to `2`` (so the gate actually drops sub-threshold windows).

The reversibility requirement: rollback = flip a flag. There are two independent flags:

- `indexing.fusion.min_crossings = 0`` ? restores prior (ungated) behavior even on the fusion path. This is the **Gate-A-specific reversal** the prompt asks for (step 2d).
- `indexing.default_segmenter = "wasb_hybrid"`` ? restores the exact prior served segmenter entirely.

The flagged risk (prompt explicitly asks me to flag this): the project's **committed** default in `models.py`` is `yolo_hybrid``, but the **live** `config.json`` default is `gemini``. The actual GPU-run default the owner uses is `wasb_hybrid`` (per current-state). The cleanest, safest path that keeps `FusionSegmenter``-default byte-identical to `wasb_hybrid`` (verified by #117) is:

- Change `FusionConfig.min_crossings`` default from `0`` ? `2`` (the validated Gate-A operating

point).

- Change `config.json`'s `indexing.default\_segmenter` from `gemini` ? `fusion\_hybrid` (the served default the owner actually wants flipped for badminton GPU runs).

I'll **keep the `models.py` code default `default\_segmenter = "yolo\_hybrid"` unchanged** (changing the hardcoded fallback would surprise non-config callers and other sports). The served flip belongs in `config.json` ? that's the operator-facing default and it's the one the prompt says drives `main.py`'s served-default selection. Flipping `min\_crossings` default in `models.py` is the right home for "Gate-A is on by default" because it's the value that makes Gate-A active everywhere `fusion\_hybrid` runs.

Wait ? let me reconsider. If I flip `FusionConfig.min\_crossings` default to `2` in `models.py`, then **any** use of `fusion\_hybrid` (including offline harness re-scoring that doesn't override it) gets gated by default. That's correct and intended ("Gate-A as DEFAULT detection behavior"). The offline harness explicitly constructs Variants with their own `min\_crossings`, so it's unaffected. And the persisted superset is never gated regardless (only `\_select\_for\_handoff` is). Good ? that preserves step 2e.

Let me verify the harness's `compare`/`build\_candidates` path doesn't read `FusionConfig.min\_crossings` for persistence. I already confirmed from `trajectory\_hybrid.py` that persistence (`add\_candidate\_segment`) uses the full `window\_stats` regardless of `\_select\_for\_handoff`. The fusion offline extractor uses `build\_candidates` separately. Let me quickly confirm the harness path doesn't accidentally inherit a gated default.

31. user (2026-06-14T10:11:24.325Z)

```
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/calibrate_local.py:7:
-> rally_gate (min_crossings)          -> rally windows
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/calibrate_local.py:34:#
(velocity_thresh, merge_gap, min_window_duration, min_crossings)
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/calibrate_local.py:52:
wins = rally_gate.apply_gate(points, wins, fps, min_crossings=mc)
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/distill_local.py:56:    gated =
rally_gate.gate_windows(points, intervals, fps, min_crossings=0) # for net-crossings
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/eval_partial_labels.py:62:
preds = rally_gate.apply_gate(points, preds, fps, min_crossings=gate_crossings)
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/experiment.py:90:    -
`min_crossings` ? decision-gate **Gate A**: keep only windows whose persisted
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/experiment.py:111:
min_crossings: int = 0
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/experiment.py:133:
"min_crossings": self.min_crossings,
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/experiment.py:204:    if
variant.min_crossings > 0:
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/experiment.py:206:        mask
= [m and (col[i] >= variant.min_crossings) for i, m in enumerate(mask)]
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/export_wasb_segments.py:142:
comparison_out: Optional[str] = None, min_crossings: int = 0) -> dict:
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/export_wasb_segments.py:145:
# the shuttle to cross the net >= min_crossings times within the window.
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/export_wasb_segments.py:146:
points = TrackNetRunner.parse_trajectory_csv(trajectory_csv) if min_crossings > 0 else None
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/export_wasb_segments.py:151:
kept = rally_gate.apply_gate(points, segs, fps, min_crossings=min_crossings)
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/export_wasb_segments.py:153:
f"(removed {len(segs) - len(kept)} with <{min_crossings} net-crossings)")
C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/export_wasb_segments.py:222:
comparison_out=args.comparison_out, min_crossings=args.min_crossings)
Binary file C:/Users/avidu/Projects/badminton-highlight-
indexer/backend/eval/__pycache__/calibrate_local.cpython-313.pyc matches
Binary file C:/Users/avidu/Projects/badminton-highlight-
indexer/backend/eval/__pycache__/distill_local.cpython-313.pyc matches
Binary file C:/Users/avidu/Projects/badminton-highlight-
indexer/backend/eval/__pycache__/eval_partial_labels.cpython-313.pyc matches
Binary file C:/Users/avidu/Projects/badminton-highlight-
```

indexer/backend/eval/__pycache__/experiment.cpython-313.pyc matches
Binary file C:/Users/avidu/Projects/badminton-highlight-
indexer/backend/eval/__pycache__/export_wasb_segments.cpython-313.pyc matches

32. assistant (2026-06-14T10:11:28.679Z)

Confirmed: the eval harness (`experiment.py`, `distill\_local.py`) uses its own explicit
`Variant.min\_crossings` / `rally\_gate` parameters and never reads `FusionConfig.min\_crossings`.
So flipping the `FusionConfig` default to `2` does **not** touch the offline ablation harness ?
it only affects the live `process\_video` served path. This keeps the measurement-honesty
separation clean.

Now let me set up the isolated worktree and create the branch from `origin/master`.

33. user (2026-06-14T10:11:29.902Z)

Preparing worktree (new branch 'feat/gate-a-default')
branch 'feat/gate-a-default' set up to track 'origin/master'.
HEAD is now at 7c6498f docs(quality): personalization Phase-0 checkpoint ? eval/safety rails +
locked structural-guard decision (#145)

34. assistant (2026-06-14T10:11:33.252Z)

Worktree created at `7c6498f` (= origin/master, latest is #145). Let me verify the worktree
state and confirm the files match what I read (they're from origin/master, slightly ahead of my
earlier reads which were from the main checkout).

35. user (2026-06-14T10:11:33.952Z)

feat/gate-a-default
7c6498f docs(quality): personalization Phase-0 checkpoint ? eval/safety rails + locked
structural-guard decision (#145)

36. assistant (2026-06-14T10:11:37.070Z)

Clean ? branch is mine, 0 commits ahead of origin/master yet. Let me verify the files in the
worktree match what I read (in case #145 etc. changed them).

37. user (2026-06-14T10:11:39.058Z)

---HEAD diff (should be empty)---

38. assistant (2026-06-14T10:11:41.897Z)

Clean tree, files match origin/master. Now let me make the changes. First, the core change in
`models.py` ? flip `FusionConfig.min\_crossings` default `0` ? `2`.

39. user (2026-06-14T10:11:42.603Z)

```
78     class FusionConfig(BaseModel):
79         """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN
P2).
80
81         Every default reproduces control behaviour: the fusion segmenter persists the
82         shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']``
is
83         true ? the person-cue features, but it does NOT gate the served handoff unless a
84         threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court
mask
85         is optional: ``court_polygon=None`` ? Gate B counts every detection (player-count-
only);
86         supplied ? off-court persons (spectators / adjacent courts) are excluded.
87         """
```

```

88     # Which trajectory substrate seeds the windows (shuttle stays primary).
89     substrate: Literal["wasb", "tracknet"] = "wasb"
90     # Windowing velocity threshold for the fusion family (the engine reads
91     # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
92     # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
93     velocity_threshold: float = 0.02
94     # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
95     # enabled ? so the default path never imports torch / touches the GPU.
96     cue_flags: dict[str, bool] = Field(default_factory=dict)
97     # Served Gate A: drop windows whose shuttle net-crossings < this from the AI
handoff.
98     # 0 = OFF (no gating; persisted candidates are always the full superset for offline
re-scoring).
99     min_crossings: int = Field(default=0, ge=0)
100     # Served Gate B: when the person cue is on, drop windows with median in-court
players
101     # < this. 0 = OFF.
102     min_players: int = Field(default=0, ge=0)
103     # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
104     court_polygon: Optional[list[list[float]]] = None
105     # Net line for the person two-sided-motion split (person features only ? the shuttle
106     # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
107     net_axis: Literal["x", "y"] = "y"
108     net_position: float = Field(default=0.5, ge=0.0, le=1.0)
109

```

40. user (2026-06-14T10:11:55.946Z)

The file C:\Users\avidu\Projects\bhi-gate\backend\config\models.py has been updated successfully. (file state is current in your context ? no need to Read it back)

41. user (2026-06-14T10:12:01.972Z)

The file C:\Users\avidu\Projects\bhi-gate\backend\config\models.py has been updated successfully. (file state is current in your context ? no need to Read it back)

42. assistant (2026-06-14T10:12:05.546Z)

Now flip the served default in `config.json` from `gemini` ? `fusion\_hybrid`. Let me also add the fusion block to config.json so the served default and its Gate-A knob are explicit/visible to operators (currently config.json has no `fusion` block, so it would rely on the model default ? that works, but making it explicit aids reversibility/discoverability).

43. user (2026-06-14T10:12:06.907Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

44. assistant (2026-06-14T10:12:10.003Z)

I read config.json from the main checkout, not the worktree. Let me read the worktree copy.

45. user (2026-06-14T10:12:10.465Z)

```

1      {
2          "sport": "badminton",
3          "ai_provider": "gemini",
4          "ai_model": "gemini-flash-latest",
5          "validation": {
6              "min_resolution_width": 1280,
7              "min_resolution_height": 720,
8              "min_fps": 29.9,
9              "min_blur_threshold": 20.0,
10             "max_scene_cuts_per_minute": 0.5,
11             "relevance": {

```

```

12         "court_green_lower": [
13             35,
14             40,
15             40
16         ],
17         "court_green_upper": [
18             85,
19             255,
20             255
21         ],
22         "court_pixels_min_ratio": 0.05
23     }
24 },
25     "indexing": {
26         "default_segmenter": "gemini",
27         "use_gpu": true,
28         "motion_threshold": 0.08,
29         "min_segment_duration": 3.0,
30         "dead_time_merge_gap": 2.0,
31         "chunk_duration": 90,
32         "chunk_overlap": 10,
33         "detector_model": "fasterrcnn_resnet50_fpn",
34         "detector_score_threshold": 0.5,
35         "yolo_velocity_threshold": 0.15,
36         "yolo_frame_skip": 3,
37         "yolo_tracker": "bytetrack.yaml",
38         "yolo_prefetch_workers": 2,
39         "min_action_window_duration": 2.0,
40         "log_yolo_candidates": true,
41         "store_yolo_candidates": true,
42         "ai_handoff_padding": 2.0,
43         "ai_max_workers": 1,
44         "tracknet": {
45             "wsl_distro": "Ubuntu",
46             "repo_dir": "~/models/TrackNetV4",
47             "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
48             "conda_env": "TrackNetV4",
49             "weights_path": "",
50             "queue_length": 5,
51             "stage_in_wsl": true,
52             "wsl_stage_dir": "~/clips",
53             "timeout_sec": 1800,
54             "velocity_threshold": 0.02
55         },
56         "wasb": {
57             "wsl_distro": "Ubuntu",
58             "repo_dir": "~/models/WASB-SBDT",
59             "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
60             "conda_env": "wasb",
61             "weights_path": "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar",
62             "sport": "badminton",
63             "wsl_stage_dir": "~/clips",
64             "timeout_sec": 1800,
65             "velocity_threshold": 0.02
66         }
67     },
68     "output": {
69         "default_highlights_name": "highlights.mp4",
70         "db_name": "sports_indexer.db"
71     },
72     "stitching": {
73         "begin_padding": 0.5,
74         "end_padding": 0.5,
75         "use_cache": false,

```

```

76         "min_shot_count": 1,
77         "watermark": {
78             "enabled": true,
79             "text": "Generated by Khelsutra.guru",
80             "position": "bottom-right",
81             "opacity": 0.85,
82             "logo_path": "",
83             "font_path": ""
84         }
85     }
86 }

```

46. assistant (2026-06-14T10:12:14.464Z)

The worktree config.json matches. Now I'll flip the default segmenter to `fusion\_hybrid` and add an explicit `fusion` block (so the served Gate-A operating point and its reversal flag are visible to operators right in config.json).

47. user (2026-06-14T10:12:16.353Z)

The file C:\Users\avidu\Projects\bhi-gateA\config.json has been updated successfully. (file state is current in your context ? no need to Read it back)

48. user (2026-06-14T10:12:23.375Z)

The file C:\Users\avidu\Projects\bhi-gateA\config.json has been updated successfully. (file state is current in your context ? no need to Read it back)

49. assistant (2026-06-14T10:12:26.565Z)

Now let me add comprehensive tests. First, update `test\_config.py`. The existing `test\_fusion\_config\_defaults\_reproduce\_control` asserts `min\_crossings == 0` ? that's now ****wrong**** by design. I'll rewrite it to assert the new Gate-A-default and add a reversibility test + a served-default test.

50. user (2026-06-14T10:12:27.235Z)

```

1     """Basic tests for the typed configuration package.
2
3     These ensure:
4     - Defaults are sensible and come from the models.
5     - File overrides work.
6     - Old config files with extra keys are tolerated.
7     - save_default_config produces loadable output.
8     """
9
10    import json
11    import tempfile
12    import os
13
14
15    from backend.config import load_config, save_default_config, AppConfig
16
17

```

51. user (2026-06-14T10:12:39.724Z)

The file C:\Users\avidu\Projects\bhi-gateA\tests\test_config.py has been updated successfully. (file state is current in your context ? no need to Read it back)

52. assistant (2026-06-14T10:12:43.907Z)

Now let me check whether any other test relies on the old `min\_crossings == 0` default. Let me search across the test suite.

53. user (2026-06-14T10:12:45.681Z)

```
tests/test_experiment.py:82:                min_crossings=0, min_players=0)
tests/test_export_wasb_segments.py:89:    r0 = ex.run(traj, fps=10.0, frame_width=100.0,
out_csv=str(tmp_path / "o0.csv"), min_crossings=0)
tests/test_fusion_hybrid.py:86:    _db, provider, _res = _run() #
min_crossings default 0 -> no gating
```

54. assistant (2026-06-14T10:12:58.371Z)

Two things to address:

1. `test\_experiment.py:82` and `test\_export\_wasb\_segments.py:89` explicitly pass `min\_crossings=0` (their own Variant/function param, unaffected by the `FusionConfig` default) ? those are correct and need no change.
2. `test\_fusion\_hybrid.py:86` (`test\_default\_hands\_off\_all\_windows`) relies on the `\_run()` default which constructs `FusionSegmenter(db, {"indexing": indexing or {}})` ? i.e. an **empty** indexing dict, NOT the `FusionConfig`-defaulted one. So that test's "default" is the bare-dict default where `fcfg.get("min\_crossings", 0)` returns 0.

This is a subtlety: `FusionSegmenter.\_select\_for\_handoff` reads `idx\_cfg.get("fusion", {}).get("min\_crossings", 0)` ? a **plain-dict** lookup that defaults to 0, NOT the Pydantic `FusionConfig` default of 2. So in the served path, the `FusionConfig` default of 2 only applies if the config is loaded through Pydantic (which `main.py` does ? `global\_config` is an `AppConfig`, and `self.config.get("indexing")` returns the model_dump that includes `fusion.min\_crossings=2`).

Let me verify this end-to-end: does `AppConfig().get("indexing")` produce a dict containing `fusion: {min\_crossings: 2}` ? This is the critical correctness check.

55. user (2026-06-14T10:12:59.315Z)

```
type(idx): dict
fusion in idx: True
fusion.min_crossings: 2
default_segmenter: yolo_hybrid
```

56. assistant (2026-06-14T10:13:06.713Z)

Confirmed. When the served path loads config via Pydantic `AppConfig`, `self.config.get("indexing")["fusion"]["min\_crossings"]` is `2` ? Gate-A fires by default in `\_select\_for\_handoff`. The bare-dict path used in the existing unit test (`{"indexing": {}}`) keeps `min\_crossings=0` because it never instantiates `FusionConfig`.

This means I need a test that exercises the **real served default** ? building the segmenter from a real `AppConfig` (not a bare dict) and asserting Gate-A drops the held-shuttle window. The existing `test\_default\_hands\_off\_all\_windows` uses a bare dict and is correct as-is (it documents the "no fusion config ? 0" fallback). But I should update its comment and add the AppConfig-driven test. Let me update the fusion test file.

First, let me look at how `\_run` constructs config and add an AppConfig-based variant.

57. user (2026-06-14T10:13:09.109Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

58. assistant (2026-06-14T10:13:13.817Z)

I need to read the worktree copy (I read the main-checkout copy earlier). Let me read it.

59. user (2026-06-14T10:13:13.995Z)

```
1      """FusionSegmenter tests (mocked runner/provider/db ? no GPU/WSL).
2
3      Proves: net_crossings is persisted; the person cue is OFF by default (no person-feature
```



```

4     keys, PersonDetector never built); the persisted candidate set is the full superset of
the
5     substrate windows; served Gate A drops sub-threshold windows from the AI handoff while
6     persistence keeps the superset; and the person cue, when enabled, adds the person
features.
7     """
8     from types import SimpleNamespace
9     from unittest.mock import MagicMock, patch
10
11     from backend.pipeline.detectors.fusion_features import PERSON_FEATURE_NAMES
12     from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter
13
14     # Window A [1,4]s: shuttle alternates across the net every frame -> many crossings (a
rally).
15     # Window B [10,12]s: shuttle sits at the net level (held) -> 0 crossings. fps=30.
16     WIN_A = {"start": 1.0, "end": 4.0, "active_frame_count": 90, "peak_velocity": 0.4,
17             "mean_velocity": 0.2, "track_id_count": 2, "chunk_index": 0}
18     WIN_B = {"start": 10.0, "end": 12.0, "active_frame_count": 60, "peak_velocity": 0.1,
19             "mean_velocity": 0.05, "track_id_count": 1, "chunk_index": 0}
20
21
22     def _points():
23         pts = []
24         for f in range(30, 120):          # window A frames: y flips 100<->900 around the net
(~500)
25             pts.append(SimpleNamespace(frame=f, visible=True, x=500.0, y=100.0 if f % 2 == 0
else 900.0))
26         for f in range(300, 360):        # window B frames: y == net level -> deadband -> no
crossings
27             pts.append(SimpleNamespace(frame=f, visible=True, x=500.0, y=500.0))
28         return pts
29
30
31     def _runner():
32         r = MagicMock()
33         r.healthcheck.return_value = (True, "env ok")
34         r.run_predict.return_value = "out.csv"
35         r.parse_trajectory_csv.return_value = _points()
36         r.trajectory_to_action_windows.return_value = [dict(WIN_A), dict(WIN_B)]
37         return r
38
39
40     def _run(indexing=None, provider_segments=None):
41         provider = MagicMock()
42         provider.analyze_video.return_value = (provider_segments or [{"start_time": 0.5,
"end_time": 2.0}], {})
43         db = MagicMock()
44         db.add_candidate_segment.side_effect = [101, 102, 103, 104]
45         db.add_play_segment.return_value = 200
46
47         with patch.object(FusionSegmenter, "_build_runner", return_value=(_runner(),
{"weights_path": "w"})), \
48             patch.object(FusionSegmenter, "_probe_fps_width", return_value=(30.0, 1280.0)), \
49             patch("backend.pipeline.segmenters.trajectory_hybrid.get_video_duration",
return_value=30.0), \
50             patch("backend.pipeline.segmenters.trajectory_hybrid.extract_video_chunk",
return_value=True), \
51             patch("backend.pipeline.segmenters.trajectory_hybrid.provider_registry") as
preg, \
52             patch("os.environ.get", return_value="dummy_key"):
53             preg.get.return_value = provider
54             seg = FusionSegmenter(db, {"indexing": indexing or {}})
55             res = seg.process_video("v1", "clip.mp4")
56             return db, provider, res

```

```

57
58
59 def _stats_calls(db):
60     """Persisted stats dicts, in call order."""
61     return [kw["stats"] for _a, kw in db.add_candidate_segment.call_args_list]
62
63
64 def test_net_crossings_persisted_and_person_off_by_default():
65     db, _provider, _res = _run()
66     stats = _stats_calls(db)
67     assert len(stats) == 2 # both windows persisted
68     (superset)
69     assert all("net_crossings" in s for s in stats)
70     # Window A (rally) crosses many times; window B (held) zero.
71     assert stats[0]["net_crossings"] >= 2
72     assert stats[1]["net_crossings"] == 0.0
73     # Person cue OFF by default -> NONE of the person features present.
74     assert all(not any(k in s for k in PERSON_FEATURE_NAMES) for s in stats)
75     # The 4 base motion keys are still there.
76     assert all({"peak_velocity", "mean_velocity", "active_frame_count",
77 "track_id_count"} <= set(s)
78         for s in stats)
79
80 def test_persisted_candidates_are_superset_of_substrate_windows():
81     db, _provider, _res = _run()
82     spans = [(kw["start_time"], kw["end_time"]) for _a, kw in
83 db.add_candidate_segment.call_args_list]
84     assert spans == [(1.0, 4.0), (10.0, 12.0)] # exactly the substrate windows,
85     unfiltered
86
87 def test_default_hands_off_all_windows():
88     _db, provider, _res = _run() # min_crossings default 0 -> no
89     gating
90     assert provider.analyze_video.call_count == 2
91
92 def test_gate_a_drops_low_crossing_window_from_handoff_only():
93     db, provider, _res = _run(indexing={"fusion": {"min_crossings": 2}})
94     # Persistence is still the full superset (offline substrate intact)...
95     assert len(_stats_calls(db)) == 2
96     # ...but only the high-crossing window A reaches the AI handoff.
97     assert provider.analyze_video.call_count == 1
98
99 def test_person_cue_adds_features_when_enabled():
100     # Two in-court players, each shifting between two sampled frames ? exercises the
101     REAL
102     # glue (in-court counts -> players_in_court; track displacement ->
103     mean_player_motion),
104     # not just key-presence.
105     fake_pd = MagicMock()
106     fake_pd.detections_per_frame.return_value = {
107         "frames": {
108             30: [(1, (10, 10, 30, 40)), (2, (60, 10, 80, 40))],
109             33: [(1, (16, 10, 36, 40)), (2, (66, 10, 86, 40))],
110         },
111         "fps": 30.0, "frame_width": 1000.0, "frame_height": 1000.0,
112     }
113     with patch("backend.pipeline.detectors.person_detector.PersonDetector",
114 return_value=fake_pd):
115         db, _provider, _res = _run(indexing={"fusion": {"cue_flags": {"person": True}}})
116         stats = _stats_calls(db)
117         assert all(all(k in s for k in PERSON_FEATURE_NAMES) for s in stats)

```

```

114         assert all("net_crossings" in s for s in stats)
115         assert stats[0]["players_in_court"] == 2.0          # 2 in-court players every
sampled frame
116         assert stats[0]["mean_player_motion"] > 0.0        # both tracks moved -> non-zero
motion
117         fake_pd.detections_per_frame.assert_called()       # the person detector WAS used
118
119
120     def test_gate_b_drops_low_player_window_from_handoff():
121         """Gate B: with the person cue on and min_players=2, a window whose in-court player
122         count is below 2 is dropped from the handoff ? but still persisted (superset)."""
123         def _side(video_path, start_f, end_f, frame_skip):
124             if start_f >= 300:          # window B [10,12]s -> start frame ~300: only 1 player
125                 frames = {start_f: [(1, (10, 10, 30, 40))]}
126             else:                      # window A [1,4]s -> start frame ~30: 2 players
127                 frames = {start_f: [(1, (10, 10, 30, 40)), (2, (60, 10, 80, 40))]}
128             return {"frames": frames, "fps": 30.0, "frame_width": 1000.0, "frame_height":
1000.0}
129
130         fake_pd = MagicMock()
131         fake_pd.detections_per_frame.side_effect = _side
132         with patch("backend.pipeline.detectors.person_detector.PersonDetector",
return_value=fake_pd):
133             db, provider, _res = _run(indexing={"fusion": {"cue_flags": {"person": True},
"min_players": 2}})
134             stats = _stats_calls(db)
135             assert len(stats) == 2          # superset persisted (both)
136             assert provider.analyze_video.call_count == 1      # only window A (>=2 players)
handed off
137             assert stats[0]["players_in_court"] >= 2 and stats[1]["players_in_court"] < 2
138
139
140     def test_build_runner_selects_substrate():
141         """_build_runner picks the WASB or TrackNet runner by indexing.fusion.substrate."""
142         seg = FusionSegmenter(MagicMock(), {"indexing": {}})
143         with patch("backend.pipeline.segmenters.fusion_hybrid.WasbRunner") as W, \
144             patch("backend.pipeline.segmenters.fusion_hybrid.WasbConfig") as WC, \
145             patch("backend.pipeline.segmenters.fusion_hybrid.TrackNetRunner") as T, \
146             patch("backend.pipeline.segmenters.fusion_hybrid.TrackNetConfig") as TC:
147             WC.from_indexing_cfg.return_value = MagicMock(weights_path="w")
148             TC.from_indexing_cfg.return_value = MagicMock(weights_path="t", queue_length=5)
149             _runner_w, extras_w = seg._build_runner({"fusion": {"substrate": "wasb"}})
150             _runner_t, extras_t = seg._build_runner({"fusion": {"substrate": "tracknet"}})
151             _runner_d, extras_d = seg._build_runner({})          # default -> wasb
152             assert W.called and extras_w["fusion_substrate"] == "wasb"
153             assert T.called and extras_t["fusion_substrate"] == "tracknet"
154             assert extras_t["queue_length"] == 5
155             assert extras_d["fusion_substrate"] == "wasb"
156

```

60. assistant (2026-06-14T10:13:18.422Z)

Now I'll add the import and a dedicated section of tests that exercise the ****served default** through a real AppConfig** (the path ``main.py`` uses). I'll add a helper ``_run_with_config`` that accepts a full config object, and tests for: (a) default AppConfig applies Gate-A; (b) held-shuttle window dropped by default; (c) real rally kept; (d) reversibility via ``min_crossings=0``; (e) persisted superset unchanged under the default gate.

61. user (2026-06-14T10:13:21.440Z)

The file `C:\Users\avidu\Projects\bhi-gateA\tests\test_fusion_hybrid.py` has been updated successfully. (file state is current in your context ? no need to Read it back)

62. assistant (2026-06-14T10:13:26.924Z)

Now refactor `\_run` to optionally take a full config object, and update the now-misleading comment on `test\_default\_hands\_off\_all\_windows` (it uses a bare dict, where the dict-default is 0 ? that's the genuine "no fusion config block at all" fallback, which I'll keep but clarify).

63. user (2026-06-14T10:13:38.001Z)

The file C:\Users\avidu\Projects\bhi-gateA\tests\test_fusion_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

64. user (2026-06-14T10:14:01.093Z)

The file C:\Users\avidu\Projects\bhi-gateA\tests\test_fusion_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

65. assistant (2026-06-14T10:14:05.225Z)

Now let me also update the module docstring to mention the new default behavior, then run the test suite for these two files.

66. user (2026-06-14T10:14:10.510Z)

The file C:\Users\avidu\Projects\bhi-gateA\tests\test_fusion_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

67. user (2026-06-14T10:14:19.866Z)

```
..... [100%]  
26 passed in 4.91s
```

68. assistant (2026-06-14T10:14:23.464Z)

All 26 pass. Now let me check whether any **other** test in the suite assumed the old served default or the old `min\_crossings=0` default ? particularly tests that load the project's `config.json` or assert the served default. Let me search broadly.

69. user (2026-06-14T10:14:24.851Z)

```
tests/test_api_endpoints.py  
tests/test_automated_provider.py  
tests/test_config.py  
tests/test_config_wasb.py  
tests/test_eval_batch.py  
tests/test_fusion_hybrid.py  
tests/test_providers.py  
tests/test_reporting.py  
tests/test_reporting_harvest.py  
tests/__pycache__/test_config.cpython-313-pytest-9.0.3.pyc  
tests/__pycache__/test_fusion_hybrid.cpython-313-pytest-9.0.3.pyc
```

70. user (2026-06-14T10:14:28.093Z)

```
tests/test_api_endpoints.py:182:      """config.json stitching.{begin,end}_padding must reach the  
ffmpeg trim boundaries.  
tests/test_reporting_harvest.py:46:      config={"ai_provider": "gemini", "ai_model": "gemini-  
flash", "indexing": {"default_segmenter": "wasb_hybrid"}},  
tests/test_reporting_harvest.py:122:      config={"ai_provider": "gemini", "indexing":  
{"default_segmenter": "gemini"}},  
tests/test_reporting_harvest.py:137:      config={"indexing": {"default_segmenter":  
"motion"}})  
tests/test_reporting_harvest.py:146:      cfg = {"ai_provider": "gemini", "indexing":  
{"default_segmenter": "wasb_hybrid"}}  
tests/test_reporting_harvest.py:189:      config={"ai_provider": "gemini", "indexing":  
{"default_segmenter": "wasb_hybrid"}},
```

tests/test_config_wasb.py:5:shuttle substrate's config) was discarded ? config.json overrides did nothing.

71. assistant (2026-06-14T10:14:31.015Z)

Those `test\_reporting\_harvest.py` references use their own inline config dicts (not the shipped `config.json`), so they're unaffected by my change. The `test\_config\_wasb.py:5` is a docstring comment. None of these depend on the shipped served default. Now let me run the full suite to confirm nothing regressed.

72. user (2026-06-14T10:15:07.339Z)

```
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
obsreport (optional reporting dep): available
=====
..... [ 12%]
..... [ 24%]
..... [ 36%]
..... [ 49%]
..... [ 61%]
..... [ 73%]
..... [ 86%]
..... [ 98%]
..... [100%]
584 passed in 33.67s
=====
RESULT: PASS - all tests green
=====
```

73. assistant (2026-06-14T10:15:10.899Z)

584 passed (was ~514 earlier, plus my new tests). Now run ruff and mypy as the CI gates do.

74. user (2026-06-14T10:15:43.551Z)

```
All checks passed!
=== MYPY ===
backend\annotations\base.py:85: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\sports\__init__.py:7: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs [annotation-unchecked]
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 93 source files
```

75. assistant (2026-06-14T10:15:46.513Z)

ruff clean, mypy clean (only the standard `annotation-unchecked` notes, no errors). Let me verify the exact CI gate commands match what I ran, by checking the CI workflow.

76. user (2026-06-14T10:15:48.221Z)

```
ci.yml
.github/workflows/ci.yml:16: # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle +
bugbear.
.github/workflows/ci.yml:17: # Also the repo-wide syntax guard ? ruff reports invalid-syntax on
every
.github/workflows/ci.yml:28: - name: Install ruff
.github/workflows/ci.yml:29:   run: python -m pip install ruff
.github/workflows/ci.yml:31:   run: ruff check . --output-format github
```

```
.github/workflows/ci.yml:33: # Type gate (mypy.ini at repo root): lenient green baseline with
an explicit
.github/workflows/ci.yml:35: # up. Typed core deps are installed so mypy sees their real
signatures
.github/workflows/ci.yml:45: - name: Install mypy + typed core deps
.github/workflows/ci.yml:46: run: python -m pip install mypy pydantic fastapi uvicorn
python-dotenv google-genai numpy
.github/workflows/ci.yml:48: run: mypy backend training
.github/workflows/ci.yml:63: run: python -m pip install pytest fastapi uvicorn pydantic
python-dotenv google-genai
.github/workflows/ci.yml:65: run: python -m pytest tests/test_import_smoke.py -v
.github/workflows/ci.yml:81: run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
.github/workflows/ci.yml:83: run: python -m pip install -r requirements.txt pytest
pytest-cov httpx # httpx: fastapi TestClient
.github/workflows/ci.yml:85: # pytest.importorskip ? run_all_tests.py prints which. The
coverage floor
.github/workflows/ci.yml:92: run: python run_all_tests.py --cov=backend --cov=training
--cov-report=term --cov-fail-under=70
```

77. assistant (2026-06-14T10:15:56.269Z)

The 4 CI lanes are: ruff (`ruff check .`), mypy (`mypy backend training`), import-smoke (`pytest test\_import\_smoke.py`), and full-suite-with-coverage (`run\_all\_tests.py --cov ... --cov-fail-under=70`). All locally green. Let me run the coverage-gated variant to confirm the 70% floor still holds with my added tests.

78. user (2026-06-14T10:16:39.075Z)

```
training\gen0\harness.py                108      11      90%
-----
TOTAL                                6259    1243      80%
Required test coverage of 70% reached. Total coverage: 80.14%
584 passed, 4 warnings in 39.57s
=====
RESULT: PASS - all tests green
=====
```

79. assistant (2026-06-14T10:16:43.158Z)

All 4 CI lanes pass locally (80% coverage, above the 70% floor). Now STEP 3 ? record the promotion. First, append a Gate-A promotion record to the leaderboard JSON. Let me construct a record consistent with the existing schema (it's a list of comparison records). I'll add a promotion-flavored record that captures the validated metrics and the served-flip decision.

80. user (2026-06-14T10:16:43.800Z)

```
88         "ties": 0.0,
89         "n_effective": 6.0,
90         "p_value": 0.6875
91     }
92 }
93 }
94 ]
```

81. assistant (2026-06-14T10:16:49.272Z)

The leaderboard entries are A/B comparison records. The Gate-A win metrics (?F1 +0.074, CI [+0.042,+0.105], 6/6, p=0.031, over-seg 1.68?1.01) came from the first-A/B archive (CONTROL vs Gate-A), not yet in this JSON file (which only has the P3/P4 fusion records). I'll append a record in the same shape capturing the CONTROL-vs-GateA comparison plus a `promotion` annotation block documenting the served default flip. Let me append it.

82. user (2026-06-14T10:16:58.733Z)

The file C:\Users\avidu\Projects\bhi-gate\eval_baselines\experiment_leaderboard.json has been updated successfully. (file state is current in your context ? no need to Read it back)

83. user (2026-06-14T10:17:02.872Z)

records: 3
last record keys: ['timestamp', 'iou', 'label_set_hash', 'num_videos', 'promotion', 'variant_a', 'variant_b', 'delta', 'honest_delta_f1']
promotion.decision: PROMOTED ? Gate A (net_crossings>=2) flipped to the live served default

84. assistant (2026-06-14T10:17:06.953Z)

Valid JSON, 3 records. Now extend the `docs/QUALITY\_ITERATIONS.md` Gate-A entry to PROMOTED-as-default with the served-flip detail. Let me read the worktree copy of that entry to edit precisely.

85. user (2026-06-14T10:17:07.677Z)

```
91      ### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125
corroboration *(2026-06-14)*
92      **Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The
measured win
93      (?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg
1.68?1.01 ? the
94      [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md))
clears the honest
95      bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the
eval-only Gate A
96      folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the
recall-oriented *proposal*
97      candidate set; flipping the **live served default** stays owner-gated until a real
fusion-path run confirms it
98      (the production fusion segmenter is P2/unverified on real video) ? so this records the
promotion + rationale,
99      not a silent default flip. **Corroborated on real footage:** offline re-score of
GX010125_1080p's *cached*
100     trajectory (no GPU, isolated `output/GX010125_run/`) ? 45 candidates ? CONTROL 45
rallies vs **Gate-A 27** ;
101     Gate-A dropped **18 low-crossing windows (~1.7 min)** of likely held-shuttle/feed FPs
(no golden labels for
102     this footage ? divergence, not F1).
103
104     **Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-
indexer/pull/130).** The
105     first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff
artifact. Added
106     additive/default-OFF `Variant.tune_threshold` (operating point chosen on the **train**
fold) + `calibrate`
107     (Platt/isotonic, C2). Validated on the n=6 golden set: **mean F1 0.307 ? 0.423
(+0.116)** , testlarge
108     0.000 ? 0.300, and the learned variant now **beats CONTROL (0.388)** . Threshold-in-LOVO
is the big lever;
109     calibration is available too.
110
111     **Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
112     ([PER_VIDEO_OUTPUT_LAYOUT.md](PER_VIDEO_OUTPUT_LAYOUT.md)) so each video's artifacts are
self-contained.
```

86. assistant (2026-06-14T10:17:12.879Z)

I'll add a new dated sub-entry above this one (newest-first, matching the doc's ordering) recording the served-default promotion, and update the in-place owner-gated caveat to point to it (keeping the history honest ? I won't rewrite the prior caveat, I'll annotate that it's now resolved).

87. user (2026-06-14T10:17:33.000Z)

The file C:\Users\avidu\Projects\bhi-gateA\docs\QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

88. assistant (2026-06-14T10:17:36.986Z)

Now let me re-run ruff/mypy/tests one more time after the doc+json changes (the JSON change could affect any test reading it), then review the full diff before committing.

89. user (2026-06-14T10:17:46.503Z)

```
backend/eval/experiment.py
backend/eval/fusion_audio.py
backend/eval/fusion_compare.py
=== rerun fusion+config+eval tests ===
..... [100%]
53 passed in 4.95s
```

90. assistant (2026-06-14T10:17:49.391Z)

No test reads the leaderboard JSON as a fixture (only `backend/eval/\*` write to it). Now let me review the complete diff before committing.

91. user (2026-06-14T10:17:50.521Z)

```
backend/config/models.py      | 33 ++++++---
config.json                   |  8 ++-
docs/QUALITY_ITERATIONS.md    | 36 ++++++---
eval_baselines/experiment_leaderboard.json | 45 ++++++
tests/test_config.py          | 29 ++++++-
tests/test_fusion_hybrid.py   | 86 ++++++
6 files changed, 215 insertions(+), 22 deletions(-)
=== verify only intended files changed ===
M backend/config/models.py
M config.json
M docs/QUALITY_ITERATIONS.md
M eval_baselines/experiment_leaderboard.json
M tests/test_config.py
M tests/test_fusion_hybrid.py
```

92. assistant (2026-06-14T10:17:55.034Z)

Exactly the 6 intended files. Let me do a final full-suite + ruff + mypy confirmation, then review the core code diff once more for correctness.

93. user (2026-06-14T10:18:34.564Z)

```
All checks passed!
=== mypy ===
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 93 source files
=== full suite ===
584 passed in 33.38s
=====
RESULT: PASS - all tests green
=====
```

94. assistant (2026-06-14T10:18:41.382Z)

All green. Let me view the core code diff (models.py + config.json) once more to confirm correctness before committing.

95. user (2026-06-14T10:18:42.646Z)

```
- Every default reproduces control behaviour: the fusion segmenter persists the
- threshold is raised (`min_crossings`/`min_players` default 0 = OFF). The court mask
+ **Gate A (the held-shuttle guard) is ON by default** (`min_crossings=2`): a window
+ is `fusion_hybrid` (config.json `indexing.default_segmenter`); the offline experiment
+ harness is unaffected (it sets `Variant.min_crossings` explicitly, never reading this).
+ **Reversible ? rollback is a single flag:** set `indexing.fusion.min_crossings: 0` to
+ restore the pre-Gate-A (ungated) handoff, or `indexing.default_segmenter: wasb_hybrid`
+ to revert the served segmenter entirely. With `min_crossings=0` the fusion segmenter
+ The person cue (Gate B) stays OFF by default: the fusion segmenter persists the shuttle
- min_crossings: int = Field(default=0, ge=0)
+ min_crossings: int = Field(default=2, ge=0)
-     "default_segmenter": "gemini",
+     "default_segmenter": "fusion_hybrid",
+     "fusion": {
+         "min_crossings": 2,
```

96. assistant (2026-06-14T10:18:46.656Z)

The two load-bearing changes are exactly right: `min\_crossings: 0?2` and `default\_segmenter: gemini?fusion\_hybrid`. Now let me re-verify branch safety before committing (per the concurrent-session protocol), then stage explicit paths and commit.

97. user (2026-06-14T10:18:49.687Z)

```
feat/gate-a-default
7c6498f docs(quality): personalization Phase-0 checkpoint ? eval/safety rails + locked
structural-guard decision (#145)
=== commits ahead of origin/master (should be none yet) ===
```

98. assistant (2026-06-14T10:18:56.052Z)

Branch is mine, HEAD == origin/master (7c6498f, #145, no sibling commit landed in the gap), 0 commits ahead. Now stage explicit paths and commit.

99. user (2026-06-14T10:18:57.896Z)

```
M backend/config/models.py
M config.json
M docs/QUALITY_ITERATIONS.md
M eval_baselines/experiment_leaderboard.json
M tests/test_config.py
M tests/test_fusion_hybrid.py
```

100. assistant (2026-06-14T10:19:02.620Z)

All 6 staged, nothing extraneous. Now commit with a here-doc message (using the Bash tool, so a POSIX here-doc ? not the PowerShell `@'...'@` form that bit a prior session).

101. user (2026-06-14T10:19:11.847Z)

```
[feat/gate-a-default bb52eb1] feat(fusion): flip Gate-A (held-shuttle guard) to the served
default ? config-reversible
6 files changed, 215 insertions(+), 22 deletions(-)
```

102. assistant (2026-06-14T10:19:14.222Z)

Committed `bb52eb1`. Let me verify the log shows only my commit ahead of origin/master, then push and create the PR.

103. user (2026-06-14T10:19:17.501Z)

```
=== commits ahead of origin/master (must be ONLY mine) ===
bb52eb1 feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible
=== push ===
remote:
remote: Create a pull request for 'feat/gate-a-default' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/feat/gate-a-
default
remote:
branch 'feat/gate-a-default' set up to track 'origin/feat/gate-a-default'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      feat/gate-a-default -> feat/gate-a-default
```

104. assistant (2026-06-14T10:19:19.593Z)

Only my commit is ahead. Pushed. Now create the PR.

105. user (2026-06-14T10:19:36.854Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/146>

106. assistant (2026-06-14T10:19:39.517Z)

PR #146 created. Now let me watch the CI lanes. Let me check the run status.

107. user (2026-06-14T10:19:41.611Z)

```
Full test suite (offline, mocked)      pending 0      https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27495789295/job/81269495080
Mypy (lenient baseline) pending 0      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27495789295/job/81269495068
Ruff lint      pending 0      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27495789295/job/81269495101
Syntax sweep + import smoke (no GPU stack)      pending 0
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495789295/job/81269495078
```

108. assistant (2026-06-14T10:19:51.118Z)

All 4 lanes are queued. Let me decide on STEP 4 (optional served regression) while CI runs.

****STEP 4 decision:**** The prompt authorizes GPU+Gemini but explicitly says: do NOT block the merge on a heavy multi-video Gemini sweep ? the offline ablation (6/6, p=0.031) + unit tests are the gate. A real served golden regression would require a full GPU+Gemini fusion-path run on at least one golden video (tens of minutes, real API spend) and the golden videos' source files / WSL pipeline. Given:

- The flip is byte-reversible via a single flag,
- `FusionSegmenter`'s identity to `wasb\_hybrid` at the gate boundary is unit-verified and was adversarially reviewed in #117,
- The gate per the owner is the offline ablation + unit tests,

I'll flag the served regression as an ****optional owner-gated follow-up**** rather than running it inline and blocking the merge. This matches the prompt's explicit instruction. I'll note it in the report (and it's already documented in the PR body and QUALITY_ITERATIONS).

Let me wait for CI to complete using a bounded poll loop.

109. user (2026-06-14T10:21:33.250Z)

```
Full test suite (offline, mocked)      pass    1m39s  https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27495789295/job/81269495080
Mypy (lenient baseline) pass    36s    https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27495789295/job/81269495068
Ruff lint      pass    8s    https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27495789295/job/81269495101
Syntax sweep + import smoke (no GPU stack)      pass    18s
```

```
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495789295/job/81269495078
=== DONE polling after 105s ===
Full test suite (offline, mocked)      pass    1m39s  https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495789295/job/81269495080
Mypy (lenient baseline) pass    36s    https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495789295/job/81269495068
Ruff lint      pass    8s    https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495789295/job/81269495101
Syntax sweep + import smoke (no GPU stack)      pass    18s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27495789295/job/81269495078
```

110. assistant (2026-06-14T10:21:37.013Z)

All 4 CI lanes **pass**. The owner approved the merge. Let me re-verify branch safety one final time (no sibling commit landed on master that would make this not mergeable cleanly), then squash-merge with branch deletion.

111. user (2026-06-14T10:21:42.920Z)

```
{"mergeState":"CLEAN","mergeable":"MERGEABLE","state":"OPEN"}
=== merge ===
failed to run git: fatal: 'master' is already used by worktree at
'C:/Users/avidu/Projects/badminton-highlight-indexer'
```

112. assistant (2026-06-14T10:21:46.666Z)

The merge to the GitHub remote succeeded, but ``gh pr merge --delete-branch`` tried to also update the local ``master`` (checked out in the main worktree) and failed at the local checkout step. Let me verify the remote merge landed.

113. user (2026-06-14T10:21:48.586Z)

```
{"mergeCommit":"570cf2b1bc2455bc235a2b6eb0d60a09c28b833a","mergedAt":"2026-06-14T10:21:42Z","state":"MERGED"}
```

114. assistant (2026-06-14T10:21:52.646Z)

The PR is **MERGED** on the remote (merge commit ``570cf2b``). The error was only the local-checkout convenience step. Let me confirm the remote branch was deleted, then clean up the worktree and prune.

115. user (2026-06-14T10:21:54.940Z)

```
=== remote branch deleted? (should be absent) ===
(empty above = deleted)
```

116. assistant (2026-06-14T10:21:58.833Z)

Remote branch was deleted by the merge. Now clean up: remove the worktree, delete the local branch, prune. I must do this from the main checkout (can't remove a worktree from inside itself), and avoid touching the main checkout's ``master`` working state.

117. user (2026-06-14T10:22:02.759Z)

```
=== delete local branch ===
Deleted branch feat/gate-a-default (was bb52eb1).
=== prune stale remote-tracking ref ===
- [deleted]      (none)    -> origin/feat/telemetry-wire-fusion-golden
- [deleted]      (none)    -> origin/fix/stitching-padding-config
7c6498f..570cf2b  master    -> origin/master
```

118. assistant (2026-06-14T10:22:05.545Z)

Cleanup complete. The worktree is removed, the local branch deleted, and `origin/master` now reflects the merge (`7c6498f..570cf2b`). Let me do a final verification that the worktree is gone and the merge content is on `origin/master`.

119. user (2026-06-14T10:22:07.965Z)

```
=== worktrees (bhi-gateA should be ABSENT) ===
C:/Users/avidu/Projects/badminton-highlight-indexer 7c6498f [master]
C:/Users/avidu/Projects/bhi-ablation                7c6498f [feat/ablation-layer]
=== Gate-A flip on origin/master? ===
    "default_segmenter": "fusion_hybrid",
    "use_gpu": true,
    min_crossings: int = Field(default=2, ge=0)
```

120. assistant (2026-06-14T10:22:15.426Z)

Everything verified. The Gate-A flip (`default\_segmenter: fusion\_hybrid` + `min\_crossings: 2`) is on `origin/master`. The `bhi-gateA` worktree is gone; an unrelated sibling worktree (`bhi-ablation`, the ablation-framework agent) remains untouched as expected. The main checkout's local `master` is still at `7c6498f` (its working tree is left as-is ? the owner can `git pull --ff-only` it when convenient).

One note: I should flag the optional served regression as a follow-up task chip since GPU+Gemini was authorized but I deliberately didn't run a heavy sweep (per the prompt's instruction not to block merge on it).

121. user (2026-06-14T10:22:28.948Z)

Noted (position 1, task_id: task_ce7f9dab). A chip is showing for the user ? they can start it in a fresh worktree with one click, or dismiss it. If this suggestion becomes stale or superseded, call dismiss_task with this task_id. Currently pending: task_ce7f9dab "Run a golden served regression on the fusion_hybrid path (GPU+Gemini)". Continue your current work.

122. assistant (2026-06-14T10:22:49.164Z)

Done. Here is the report.

Gate-A default-flip ? IMPLEMENTED and MERGED

```
**PR:** #146 ? https://github.com/avidullu/badminton-highlight-indexer/pull/146
**Merge status:** MERGED to `master` (squash, merge commit `570cf2b`); remote branch deleted,
worktree removed, local branch deleted, refs pruned. The flip is confirmed live on
`origin/master`.
```

The exact change (minimal, config-reversible) ? 6 files, +215/?22

The validated insight: `**Gate-A only fires inside `FusionSegmenter._select_for_handoff`**, so making it the default requires both (1) the served segmenter to be `fusion_hybrid` and (2) `min_crossings >= 2`. Two load-bearing edits:`

- `backend/config/models.py` ? `FusionConfig.min\_crossings` default `**0 ? 2**` (Gate A active wherever `fusion\_hybrid` serves; verified
- `AppConfig().get("indexing")["fusion"]["min\_crossings"] == 2` reaches the served path).
- `config.json` ? `indexing.default\_segmenter` `**gemini ? fusion_hybrid**` + an explicit `indexing.fusion` block exposing the operating point.
- Plus the two record files (step 3) and the two test files.

`**Reversibility (rollback = one flag, no code change):** `indexing.fusion.min_crossings: 0` (Gate-A off, reproduces the substrate byte-for-byte) or `indexing.default_segmenter: wasb_hybrid` (full revert).`

Tests (all the required assertions a?e) ? `tests/test\_fusion\_hybrid.py` + `tests/test\_config.py`

- (a) default `AppConfig` applies Gate-A ?
- `test\_appconfig\_default\_applies\_gate\_a\_drops\_held\_shuttle`,
- `test\_fusion\_config\_gate\_a\_on\_by\_default`, `test\_served\_default\_segmenter\_is\_fusion\_hybrid`

- (b) held-shuttle window (``net_crossings<2``) DROPPED by default ? same test (handoff count drops to 1)
- (c) real rally (``>=2``) KEPT ? ``test_appconfig_gate_a_keeps_real_rally``
- (d) reversibility (``min_crossings=0`` restores prior behavior) ? ``test_gate_a_reversible_via_min_crossings_zero``, ``test_fusion_config_gate_a_reversible_to_zero``
- (e) persisted candidate SUPERSET unchanged ? ``test_gate_a_default_persists_full_superset``
- Rewrote the now-stale ``test_fusion_config_defaults_reproduce_control`` (asserted ``min_crossings==0``) and clarified the bare-dict ``test_no_fusion_block_hands_off_all_windows``.

Results: full suite **584** passed; ruff clean; mypy clean (no errors); coverage **80%** (?70 floor). All **4** CI lanes passed on the PR.

Records (step 3, only I touched these)

- ``docs/QUALITY_ITERATIONS.md`` ? new dated "Gate-A flipped to the SERVED DEFAULT (owner-approved)" entry with the served-flip detail; annotated the prior owner-gated caveat as now-resolved.
- ``eval_baselines/experiment_leaderboard.json`` ? appended a ``promotion`` record (?F1 +0.074, CI, 6/6, p=0.031, over-seg 1.68?1.01, served-change + rollback notes). JSON validated.

Regression (step 4)

I did **not** run a heavy GPU+Gemini served regression ? per your instruction it does not gate the merge (the offline 6/6 + p=0.031 ablation + unit tests are the gate). I flagged it as an **optional owner-gated follow-up** (background-task chip spawned).

Flagged risk

The served ``FusionSegmenter`` path is **P2** / only mock-tested on real video ? its byte-identity to ``wasb_hybrid`` at the gate boundary was verified by #117's adversarial review and unit tests, but no real Gemini-confirmed fusion-path run has executed. Mitigation: the flip is **byte-reversible** behind a single flag. A golden **served** regression on the fusion path (re-score the cached WASB CSV to avoid re-running WASB) remains the recommended follow-up to add served-side confidence.

Note: the main checkout's local ``master`` is still at ``7c6498f`` (its working tree was left untouched, as it's a shared checkout with sibling worktrees); a ``git pull --ff-only`` there will fast-forward it to ``570cf2b``.